CHAPTER 10

DYNAMICS OF PARALLEL MANIPULATORS

10.1 INTRODUCTION

In this chapter we investigate the dynamics of parallel manipulators such as the VARIAX machining center developed by Giddings & Lewis, shown in Fig. 10.1. While the kinematics of parallel manipulators have been studied extensively during the last two decades, fewer papers can be found on the dynamics of parallel manipulators. The dynamical analysis of parallel manipulators is complicated by the existence of multiple closed-loop chains. Several approaches have been proposed, including the Newton–Euler formulation (Do and Yang, 1988; Guglielmetti and Longchamp, 1994; Tsai and Kohli, 1990), the Lagrangian formulation (Lebret et al., 1993; Miller and Clavel, 1992; Pang and Shahingpoor, 1994), and the principle of virtual work (Codourey and Burdet, 1997; Miller, 1995; Tsai, 1998; Wang and Gosselin, 1997; Zhang and Song, 1993). Other approaches have also been suggested (Baiges and Duffy, 1996; Freeman and Tesar, 1988; Luh and Zheng, 1985; Murray and Lovell, 1989; Sugimoto, 1987).

The traditional Newton–Euler formulation requires the equations of motion to be written once for each body of a manipulator, which inevitably leads to a large number of equations and results in poor computational efficiency. The Lagrangian formulation eliminates all of the unwanted reaction forces and moments at the outset. It is more efficient than the Newton–Euler formulation. However, because of the numerous constraints imposed by closed loops of a parallel manipulator, deriving explicit equations of motion in terms

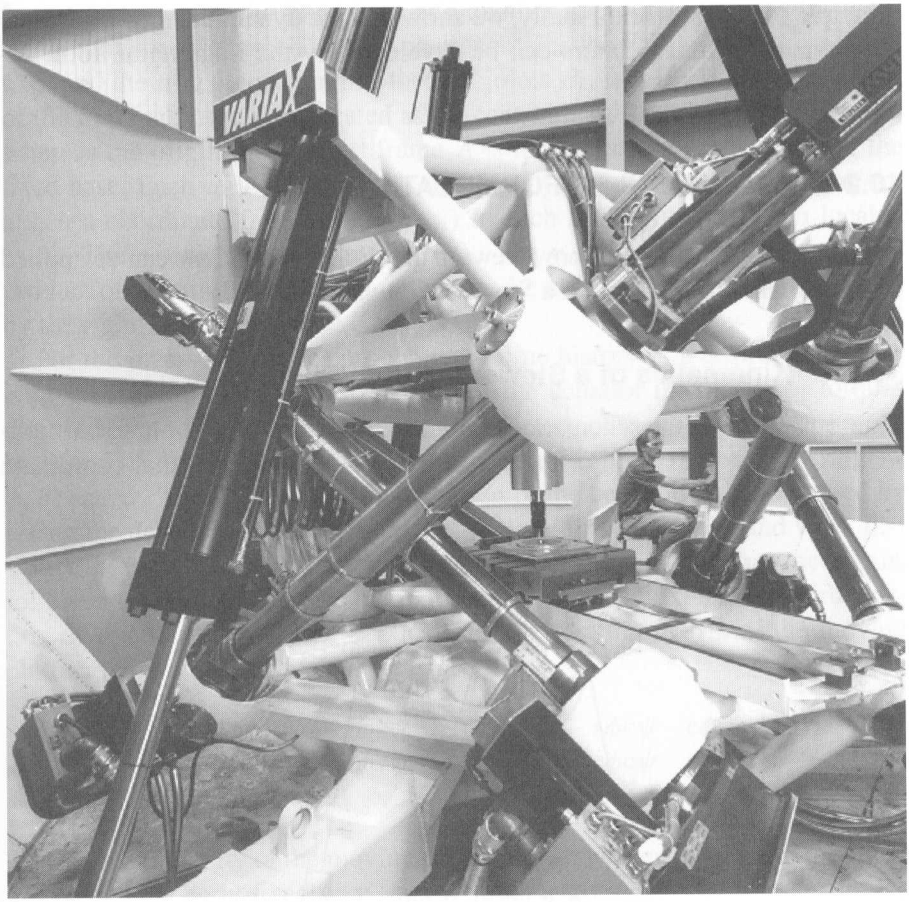

FIGURE 10.1. VARIAX® machining center. (Courtesy of Giddings & Lewis Machine Tools, Fond du Lac, Wisconsin.)

of a set of independent generalized coordinates becomes a prohibitive task. To simplify the problem, additional coordinates along with a set of Lagrangian multipliers are often introduced. In some cases, limbs are approximated by point masses by arguing that such approximation does not introduce significant modeling errors (Lebret et al., 1993; Pfreundschuh et al., 1994; Stamper, 1997; Stamper and Tsai, 1998). In this regard, the principle of virtual work appears to be the most efficient method of analysis. A comparison study of the inverse dynamics of manipulators with closed-loop geometry can be found in the work of Lin and Song (1990).

In this chapter we first describe the Newton–Euler method of analysis for parallel manipulators. Then we present a more efficient technique using the principle of virtual work. Finally, we show that the dynamical model of some simple parallel manipulators can be developed by the Lagrangian formulation.

10.2 NEWTON–EULER FORMULATION

In this section we illustrate how Newton and Euler's equations can be applied for the dynamical analysis of a Stewart–Gough platform.

10.2.1 Kinematics of a Stewart–Gough Platform

Figure 10.2 shows the schematic diagram of a Stewart–Gough platform, studied in Chapters 3 and 5. For the purpose of analysis, we attach a coordinate

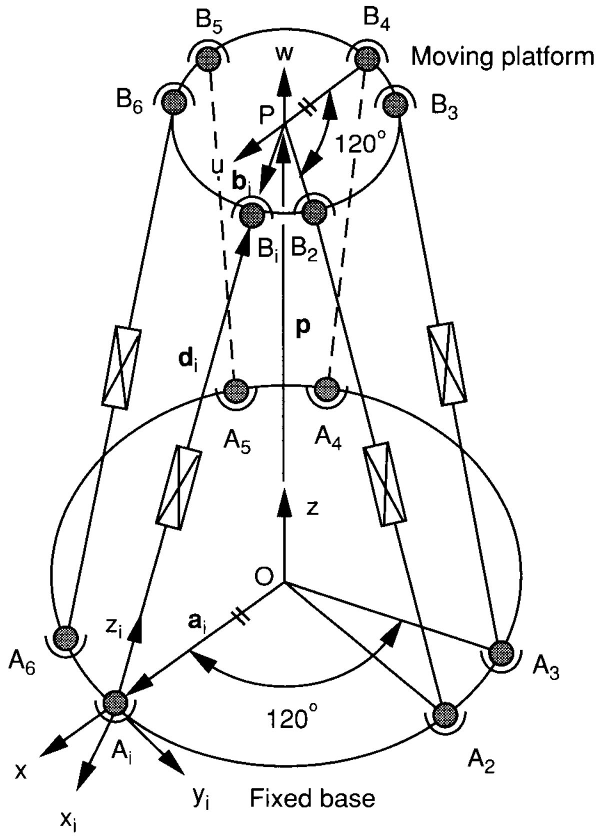

FIGURE 10.2. Schematic diagram of a Stewart–Gough platform.

frame $A(x, y, z)$ to the fixed base and another coordinate frame $B(u, v, w)$ to the moving platform. The x–y plane contains the ball joints A_i, for $i = 1$ to 6, and the u–v plane contains the ball joints B_i, for $i = 1$ to 6. The origin of the moving frame B is located at the centroid, P, of the moving platform, whereas the origin of the fixed frame A is located at the centroid, O, of the fixed base. Each of the six limbs is denoted by a vector $\mathbf{d}_i$. Furthermore, we attach a coordinate frame $C(x_i, y_i, z_i)$ to each limb with the origin located at A_i. The z_i-axis points from A_i to B_i. The y_i-axis is parallel to the cross product of two unit vectors along the z_i and z axes, and the x_i-axis is defined by the right-hand rule.

For the inverse dynamics problem, the time history of a desired trajectory is given and the problem is to determine the actuator forces and/or torques required to produce that motion. The time history of the moving platform can be described by a position vector of the centroid, $\mathbf{p}$, and three Euler angles, ϕ, θ, and ψ. The velocity and acceleration of the centroid P are obtained by taking the derivatives of $\mathbf{p}$ with respect to time; that is, $\mathbf{v}_p = \dot{\mathbf{p}}$ and $\dot{\mathbf{v}}_p = \ddot{\mathbf{p}}$.

Let the three Euler angles be defined as a rotation of ϕ about the z-axis, followed by a second rotation of θ about the rotated v-axis, and a third rotation of ψ about the rotated w-axis. Then the rotation matrix of the moving platform relative to the fixed base is given by

$$ {}^A R_B = \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix}. \tag{10.1} $$

The angular velocity of the moving platform, $\boldsymbol{\omega}_p$, written in terms of the Euler angles and the body-fixed $\mathbf{w}$, $\mathbf{v}'$, and $\mathbf{w}''$ unit vectors, is

$$ \boldsymbol{\omega}_p = \dot{\phi}\mathbf{w} + \dot{\theta}\mathbf{v}' + \dot{\psi}\mathbf{w}''. \tag{10.2} $$

In Eq. (10.2), $\mathbf{w}$, $\mathbf{v}'$, and $\mathbf{w}''$ are not orthogonal. Expressing $\mathbf{w}$, $\mathbf{v}'$, and $\mathbf{w}''$ in the fixed reference frame A, we obtain

$$ \boldsymbol{\omega}_p = \begin{bmatrix} \dot{\psi} c\phi s\theta - \dot{\theta} s\phi \\ \dot{\psi} s\phi s\theta + \dot{\theta} c\phi \\ \dot{\psi} c\theta + \dot{\phi} \end{bmatrix}. \tag{10.3} $$

The angular acceleration of the moving platform is obtained by taking the derivative of Eq. (10.3) with respect to time:

$$ \dot{\boldsymbol{\omega}}_p = \begin{bmatrix} \ddot{\psi} c\phi s\theta - \dot{\psi}\dot{\phi} s\phi s\theta + \dot{\psi}\dot{\theta} c\phi c\theta - \ddot{\theta} s\phi - \dot{\theta}\dot{\phi} c\phi \\ \ddot{\psi} s\phi s\theta + \dot{\psi}\dot{\phi} c\phi s\theta + \dot{\psi}\dot{\theta} s\phi c\theta + \ddot{\theta} c\phi - \dot{\theta}\dot{\phi} s\phi \\ \ddot{\psi} c\theta - \dot{\psi}\dot{\theta} s\theta + \ddot{\phi} \end{bmatrix}. \tag{10.4} $$

Both $\boldsymbol{\omega}_p$ and $\dot{\boldsymbol{\omega}}_p$ in Eqs. (10.3) and (10.4) are expressed in the fixed frame A. These two vectors can be transformed into the moving frame B by pre-multiplying ${}^A R_B^{\mathrm{T}}$.

(a) Position Analysis. First, we derive the location of each limb in terms of the location of the moving platform. Referring to Fig. 10.2, a vector-loop equation can be written as

$$\mathbf{a}_i + d_i \mathbf{s}_i = \mathbf{p} + \mathbf{b}_i, \tag{10.5}$$

where $\mathbf{a}_i = [a_{ix}, a_{iy}, 0]^{\mathrm{T}}$ denotes the position vector of A_i with respect to the fixed frame, ${}^B\mathbf{b}_i = [b_{iu}, b_{iv}, 0]^{\mathrm{T}}$ denotes the position vector of B_i with respect to the moving frame, $\mathbf{b}_i = [b_{ix}, b_{iy}, b_{iz}]^{\mathrm{T}}$ represents the vector ${}^B\mathbf{b}_i$ expressed in the fixed frame (i.e., $\mathbf{b}_i = {}^A R_B\, {}^B\mathbf{b}_i$), $\mathbf{s}_i$ is a unit vector pointing from A_i to B_i, and d_i is the length of limb i. Solving Eq. (10.5) for $\mathbf{s}_i$, we obtain

$$\mathbf{s}_i = \frac{\mathbf{p} + \mathbf{b}_i - \mathbf{a}_i}{d_i}, \tag{10.6}$$

where

$$d_i = |\mathbf{p} + \mathbf{b}_i - \mathbf{a}_i|. \tag{10.7}$$

Assuming that each limb is connected to the fixed base by a universal joint such that it cannot rotate about the longitudinal axis, the orientation of limb i with respect to the fixed base can be described by two Euler angles, namely a rotation of ϕ_i about the z_i-axis resulting in a (x_i', y_i', z_i') system, followed by another rotation of θ_i about the rotated y_i'-axis as shown in Fig. 10.3. Hence the rotation matrix of the ith limb can be written as

$${}^A R_i = \begin{bmatrix} c\phi_i c\theta_i & -s\phi_i & c\phi_i s\theta_i \\ s\phi_i c\theta_i & c\phi_i & s\phi_i s\theta_i \\ -s\theta_i & 0 & c\theta_i \end{bmatrix}. \tag{10.8}$$

The unit vector $\mathbf{s}_i$ expressed in the ith limb frame is given by

$${}^i\mathbf{s}_i = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \tag{10.9}$$

Upon substitution of ${}^i\mathbf{s}_i$ into $\mathbf{s}_i = {}^A R_i\, {}^i\mathbf{s}_i$, we obtain

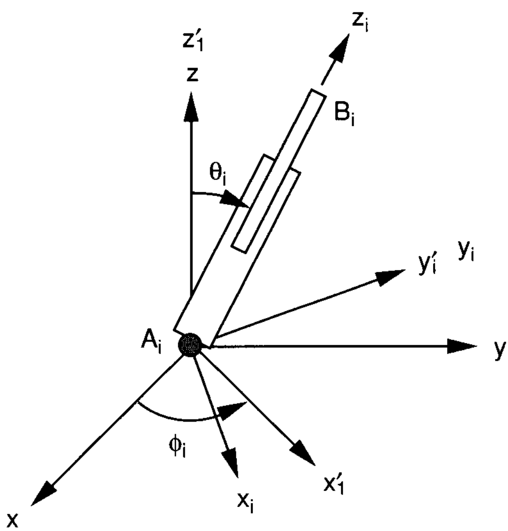

FIGURE 10.3. Euler angles of a limb.

$$\mathbf{s}_i = \begin{bmatrix} c\phi_i s\theta_i \\ s\phi_i s\theta_i \\ c\theta_i \end{bmatrix}. \tag{10.10}$$

Solving Eq. (10.10) for θ_i and ϕ_i yields

$$\begin{aligned} c\theta_i &= s_{iz}, \\ s\theta_i &= \sqrt{s_{ix}^2 + s_{iy}^2} \quad (0 \le \theta \le \pi), \\ s\phi_i &= s_{iy}/s\theta_i, \\ c\phi_i &= s_{ix}/s\theta_i, \end{aligned} \tag{10.11}$$

where s_{ix}, s_{iy}, and s_{iz} are the x, y, and z components of $\mathbf{s}_i$. Equations (10.6) and (10.11) together determine the direction and Euler angles of the ith limb in terms of the moving platform location.

As shown in Fig. 10.4, each limb consists of a cylinder (link 1) and a piston (link 2). Let e_1 be the distance between A_i and the center of mass of the ith cylinder, and let e_2 be the distance between B_i and the center of mass of the ith piston. Then the position vectors of the centers of mass of the ith cylinder and piston, $\mathbf{r}_{1i}$ and $\mathbf{r}_{2i}$, can be written as

$$\mathbf{r}_{1i} = \mathbf{a}_i + e_1\mathbf{s}_i \tag{10.12}$$

$$\mathbf{r}_{2i} = \mathbf{a}_i + (d_i - e_2)\mathbf{s}_i. \tag{10.13}$$

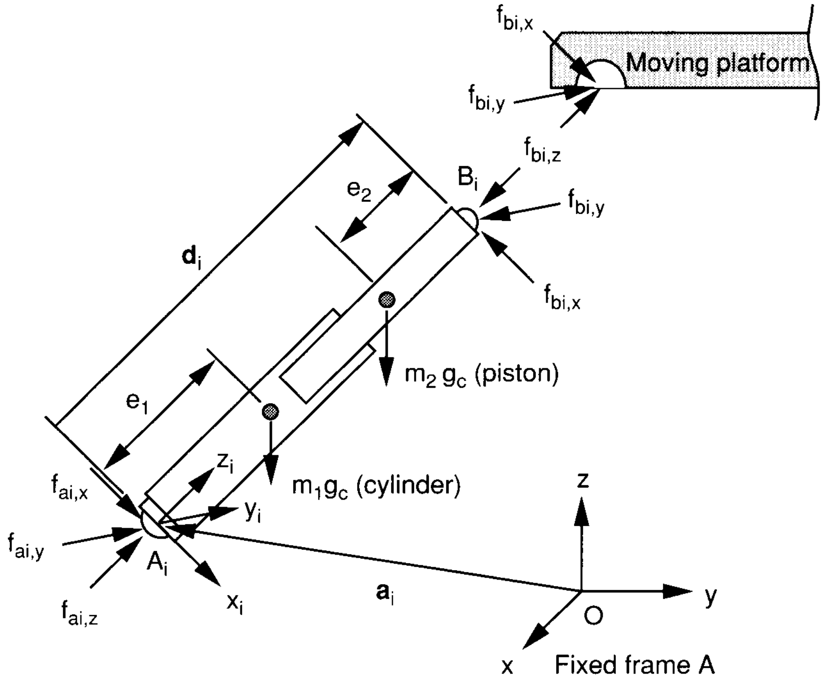

FIGURE 10.4. Free-body diagram of a typical limb.

(b) Velocity Analysis. Next, we compute the linear and angular velocities of each limb in terms of the velocity and angular velocity of the moving platform. The velocity of a ball point B_i, denoted as $\mathbf{v}_{bi}$, is found by taking the time derivative of the right-hand side of Eq. (10.5):

$$\mathbf{v}_{bi} = \mathbf{v}_p + \boldsymbol{\omega}_p \times \mathbf{b}_i. \tag{10.14}$$

Transforming $\mathbf{v}_{bi}$ to the ith limb frame yields

$$^i\mathbf{v}_{bi} = {}^iR_A \mathbf{v}_{bi}, \tag{10.15}$$

where $^i\mathbf{v}_{bi} = [\,^iv_{bix},\ ^iv_{biy},\ ^iv_{biz}]^{\mathrm{T}}$ denotes the the velocity of B_i expressed in the ith limb frame, and $^iR_A = {}^AR_i^{\mathrm{T}}$.

The velocity of B_i can also be written in terms of the angular velocity of the ith limb by taking the derivative of the left-hand side of Eq. (10.5) with respect to time:

$$^i\mathbf{v}_{bi} = d_i\, {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i + \dot{d}_i\, {}^i\mathbf{s}_i. \tag{10.16}$$

Dot-multiplying both sides of Eq. (10.16) by ${}^i\mathbf{s}_i$ yields

$$\dot{d}_i = {}^i v_{biz}. \tag{10.17}$$

Since each limb does not spin about its longitudinal axis, $\boldsymbol{\omega}_i^{\mathrm{T}}\mathbf{s}_i = 0$. Cross-multiplying both sides of Eq. (10.16) by $\mathbf{s}_i$, we obtain the angular velocity of limb i:

$$ {}^i\boldsymbol{\omega}_i = \frac{1}{d_i}({}^i\mathbf{s}_i \times {}^i\mathbf{v}_{bi}) = \frac{1}{d_i}\begin{bmatrix} -{}^i v_{biy} \\ {}^i v_{bix} \\ 0 \end{bmatrix}. \tag{10.18}$$

Once the angular velocity of the ith limb is found, the velocities of the centers of mass of the ith cylinder and piston, ${}^i\mathbf{v}_{1i}$ and ${}^i\mathbf{v}_{2i}$, are found by differentiating Eqs. (10.12) and (10.13) with respect to time:

$$ {}^i\mathbf{v}_{1i} = e_1\, {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i = \frac{e_1}{d_i}\begin{bmatrix} {}^i v_{bix} \\ {}^i v_{biy} \\ 0 \end{bmatrix} \tag{10.19}$$

$$ {}^i\mathbf{v}_{2i} = (d_i - e_2)\, {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i + \dot{d}_i\, {}^i\mathbf{s}_i = \frac{1}{d_i}\begin{bmatrix} (d_i - e_2)\, {}^i v_{bix} \\ (d_i - e_2)\, {}^i v_{biy} \\ d_i\, {}^i v_{biz} \end{bmatrix}. \tag{10.20}$$

(c) Acceleration Analysis. The acceleration of the ball point B_i, expressed in the fixed frame, is found by taking the time derivative of Eq. (10.14):

$$\dot{\mathbf{v}}_{bi} = \dot{\mathbf{v}}_p + \dot{\boldsymbol{\omega}}_p \times \mathbf{b}_i + \boldsymbol{\omega}_p \times (\boldsymbol{\omega}_p \times \mathbf{b}_i). \tag{10.21}$$

Expressing $\dot{\mathbf{v}}_{bi}$ in the ith limb frame gives

$$ {}^i\dot{\mathbf{v}}_{bi} = {}^iR_A\, \dot{\mathbf{v}}_{bi}. \tag{10.22}$$

The acceleration of B_i can also be expressed in terms of the angular acceleration of the ith limb by taking the derivative of Eq. (10.16) with respect to time:

$$ {}^i\dot{\mathbf{v}}_{bi} = \ddot{d}_i\, {}^i\mathbf{s}_i + d_i\, {}^i\dot{\boldsymbol{\omega}}_i \times {}^i\mathbf{s}_i + d_i\, {}^i\boldsymbol{\omega}_i \times ({}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i) + 2\dot{d}_i\, {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i. \tag{10.23}$$

Since each limb does not spin about its own axis, ${}^i\dot{\omega}_{iz} = 0$. Dot-multiplying both sides of Eq. (10.23) by ${}^i\mathbf{s}_i$, we obtain

$$\ddot{d}_i = {}^i\dot{v}_{biz} + d_i\, {}^i\boldsymbol{\omega}_i^2 = {}^i\dot{v}_{biz} + \frac{{}^i v_{bix}^2 + {}^i v_{biy}^2}{d_i}. \tag{10.24}$$

Cross-multiplying both sides of Eq. (10.23) by ${}^i\mathbf{s}_i$, we obtain the angular acceleration of limb i:

$$ {}^i\dot{\boldsymbol{\omega}}_i = \frac{1}{d_i}\,{}^i\mathbf{s}_i \times {}^i\dot{\mathbf{v}}_{bi} - \frac{2\dot{d}_i}{d_i}\,{}^i\boldsymbol{\omega}_i = \frac{1}{d_i}\begin{bmatrix} -{}^i\dot{v}_{biy} + \dfrac{2{}^i v_{biz}\,{}^i v_{biy}}{d_i} \\ {}^i\dot{v}_{bix} - \dfrac{2{}^i v_{biz}\,{}^i v_{bix}}{d_i} \\ 0 \end{bmatrix}. \tag{10.25} $$

Once the angular acceleration of the ith limb is found, the accelerations of the centers of mass of the ith cylinder and piston are obtained by differentiating Eqs. (10.19) and (10.20) with respect to time:

$$ \begin{aligned} {}^i\dot{\mathbf{v}}_{1i} &= e_1\,{}^i\dot{\boldsymbol{\omega}}_i \times {}^i\mathbf{s}_i + e_1\,{}^i\boldsymbol{\omega}_i \times ({}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i) \\ &= \frac{e_1}{d_i}\begin{bmatrix} {}^i\dot{v}_{bix} - \dfrac{2\,{}^i v_{biz}\,{}^i v_{bix}}{d_i} \\ {}^i\dot{v}_{biy} - \dfrac{2\,{}^i v_{biz}\,{}^i v_{biy}}{d_i} \\ -\dfrac{{}^i v_{bix}^2 + {}^i v_{biy}^2}{d_i} \end{bmatrix}, \end{aligned} \tag{10.26} $$

$$ \begin{aligned} {}^i\dot{\mathbf{v}}_{2i} &= \ddot{d}_i\,{}^i\mathbf{s}_i + (d_i - e_2)\,{}^i\dot{\boldsymbol{\omega}}_i \times {}^i\mathbf{s}_i + (d_i - e_2){}^i\boldsymbol{\omega}_i \times \left({}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i\right) \\ &\quad + 2\dot{d}_i\,{}^i\boldsymbol{\omega}_i \times {}^i\mathbf{s}_i \\ &= \frac{1}{d_i}\begin{bmatrix} (d_i - e_2)\,{}^i\dot{v}_{bix} + \dfrac{2e_2\,{}^i v_{biz}\,{}^i v_{bix}}{d_i} \\ (d_i - e_2)\,{}^i\dot{v}_{biy} + \dfrac{2e_2\,{}^i v_{biz}\,{}^i v_{biy}}{d_i} \\ d_i\,{}^i\dot{v}_{biz} + \dfrac{e_2({}^i v_{bix}^2 + {}^i v_{biy}^2)}{d_i} \end{bmatrix}. \end{aligned} \tag{10.27} $$

10.2.2 Dynamics of the Limbs

We are now in a position to perform dynamical analysis of the manipulator. To simplify the analysis, we decompose the manipulator into a moving platform and six open-loop chains (limbs). The free-body diagram of a typical limb is shown in Fig. 10.4. We combine the cylinder and piston of each limb into a subsystem and formulate the dynamical equations directly for the subsystem. In this way, reaction forces and moments between the cylinder and piston will not enter into the equations of motion. Euler's equation of motion, written about point A_i, is

$$ {}^i\mathbf{n}_i^A = \frac{d}{dt}\left({}^i\mathbf{h}_i^A\right), \tag{10.28}$$

where ${}^i\mathbf{n}_i^A$ denotes the resultant moment exerted on the ith limb about point A_i and ${}^i\mathbf{h}_i^A$ denotes combined angular momentum of the ith cylinder and piston about the same point. Both vectors are expressed in the ith limb frame.

The combined angular momentum of the ith cylinder and piston about point A_i is given by

$$ {}^i\mathbf{h}_i^A = m_1 e_1({}^i\mathbf{s}_i \times {}^i\mathbf{v}_{1i}) + m_2(d_i - e_2)({}^i\mathbf{s}_i \times {}^i\mathbf{v}_{2i}) + {}^i\mathbf{h}_{1i}^C + {}^i\mathbf{h}_{2i}^C, \tag{10.29}$$

where

$$\begin{aligned} {}^i\mathbf{h}_{1i}^C &= {}^iI_{1i}\,{}^i\boldsymbol{\omega}_i, \\ {}^i\mathbf{h}_{2i}^C &= {}^iI_{2i}\,{}^i\boldsymbol{\omega}_i \end{aligned}$$

are the angular momentums of the cylinder and piston about their respective centers of mass, and where ${}^iI_{1i}$ and ${}^iI_{2i}$ are the inertia matrices of the cylinder and piston about their respective centers of mass and expressed in the ith limb frame. Differentiating Eq. (10.29) with respect to time, we obtain

$$\begin{aligned} \frac{d}{dt}\left({}^i\mathbf{h}_i^A\right) &= m_1 e_1({}^i\mathbf{s}_i \times {}^i\dot{\mathbf{v}}_{1i}) + m_2(d_i - e_2)({}^i\mathbf{s}_i \times {}^i\dot{\mathbf{v}}_{2i}) + {}^iI_{1i}\,{}^i\dot{\boldsymbol{\omega}}_i \\ &\quad + {}^i\boldsymbol{\omega}_i \times ({}^iI_{1i}\,{}^i\boldsymbol{\omega}_i) + {}^iI_{2i}\,{}^i\dot{\boldsymbol{\omega}}_i + {}^i\boldsymbol{\omega}_i \times ({}^iI_{2i}\,{}^i\boldsymbol{\omega}_i). \end{aligned} \tag{10.30}$$

The external moment exerted on limb i about A_i is due to the gravitational force exerted at the centers of mass of the two links and the reaction forces exerted at the ball joint B_i. Since the acceleration of gravity $\mathbf{g}$ is defined in the fixed frame, it should be transformed into the limb frame. Referring to Fig. 10.4, let ${}^i\mathbf{f}_{bi} = [{}^if_{bix}, {}^if_{biy}, {}^if_{biz}]^{\mathrm{T}}$ be the force acting on the moving platform by the ith limb and ${}^A\mathbf{g} = [0, 0, -g_c]^{\mathrm{T}}$ be the acceleration of gravity. Then the resulting moment acting on the ith limb about point A_i is given by

$$\begin{aligned} {}^i\mathbf{n}_i^A &= d_i\,{}^i\mathbf{s}_i \times (-{}^i\mathbf{f}_{bi}) + [m_1e_1 + m_2(d_i - e_2)]({}^i\mathbf{s}_i \times {}^iR_A\,{}^A\mathbf{g}) \\ &= \begin{bmatrix} d_i\,{}^if_{biy} \\ -d_i\,{}^if_{bix} + m_1e_1g_c s\theta_i + m_2(d_i - e_2)g_c s\theta_i \\ 0 \end{bmatrix}. \end{aligned} \tag{10.31}$$

The dynamical equations of motion for the ith limb are obtained by substituting Eqs. (10.30) and (10.31) into (10.28). These equations of motion can be further simplified by assuming that the limbs are made up of slender cylindrical rods. Substituting Eqs. (10.30) and (10.31) into (10.28) and making use of the fact that the products of inertia are all equal to zero, I_{iz} is negligibly small, and ${}^i\omega_{iz} = 0$, we obtain

$$
\begin{aligned}
{}^i f_{bix} &= \frac{1}{d_i}[m_1 e_1 g_c \mathrm{s}\theta_i + m_2(d_i - e_2) g_c \mathrm{s}\theta_i - m_1 e_1 \, {}^i\dot{v}_{1ix} \\
&\quad - m_2(d_i - e_2)\, {}^i\dot{v}_{2ix} - I_{1iy}\, {}^i\dot{\omega}_{iy} - I_{2iy}\, {}^i\dot{\omega}_{iy}], \qquad (10.32) \\
{}^i f_{biy} &= \frac{1}{d_i}[-m_1 e_1 \, {}^i\dot{v}_{1iy} - m_2(d_i - e_2)\, {}^i\dot{v}_{2iy} \\
&\quad + I_{1ix}\, {}^i\dot{\omega}_{ix} + I_{2ix}\, {}^i\dot{\omega}_{ix}], \qquad (10.33)
\end{aligned}
$$

where I_{jix} and I_{jiy} denote the x and y components of the principal moments of inertia of the cylinder ($j = 1$) or piston ($j = 2$) about their respective centers of mass and expressed in the ith limb frame. Equations (10.32) and (10.33) determine ${}^i f_{bix}$ and ${}^i f_{biy}$ for each limb in terms of its inertia forces and moments.

10.2.3 Dynamics of the Moving Platform

In this section we formulate the dynamical equations of motion of the moving platform. These equations of motion are expressed either in the fixed frame A or in the moving frame B. Since the reaction forces, ${}^i\mathbf{f}_{bi}$, obtained in the preceding section are expressed in the ith limb frame, they should be transformed into either the fixed frame A or the moving frame B before they are substituted into the equations of motion.

First, we apply Newton's equation of motion to the moving platform and express the resulting equation in the fixed frame:

$$\sum_{i=1}^{6} {}^A\mathbf{f}_{bi} + m_p \, {}^A\mathbf{g} = m_p \, {}^A\dot{\mathbf{v}}_p, \qquad (10.34)$$

where

$${}^A\mathbf{f}_{bi} = {}^A R_i \, {}^i\mathbf{f}_{bi} \qquad (10.35)$$

denotes the reaction force exerted on the moving platform by the ith limb at a ball joint B_i and expressed in the fixed frame A. Substituting Eq. (10.8) into (10.35) and then the resulting equation into (10.34), we obtain

$$\sum_{i=1}^{6}({}^i f_{bix} \mathrm{c}\phi_i \mathrm{c}\theta_i - {}^i f_{biy} \mathrm{s}\phi_i + {}^i f_{biz} \mathrm{c}\phi_i \mathrm{s}\theta_i) = m_p \dot{v}_{px}, \qquad (10.36)$$

$$\sum_{i=1}^{6}({}^i f_{bix} s\phi_i c\theta_i + {}^i f_{biy} c\phi_i + {}^i f_{biz} s\phi_i s\theta_i) = m_p \dot{v}_{py}, \tag{10.37}$$

$$\sum_{i=1}^{6}(-{}^i f_{bix} s\theta_i + {}^i f_{biz} c\theta_i) = m_p \dot{v}_{pz} + m_p g_c. \tag{10.38}$$

The resulting moment, ${}^B\mathbf{n}_p$, taken about the center of mass of the moving platform and expressed in the moving frame B is

$${}^B\mathbf{n}_p = \sum_{i=1}^{6} {}^B\mathbf{b}_i \times {}^B\mathbf{f}_{bi}, \tag{10.39}$$

where

$${}^B\mathbf{f}_{bi} = {}^B R_A\, {}^A\mathbf{f}_{bi} = {}^B R_i\, {}^i\mathbf{f}_{bi} \tag{10.40}$$

denotes the reaction force exerted on the moving platform by the ith limb at the ball joint B_i and expressed in the moving frame B.

Assuming that u, v, and w are the principal axes of the moving platform, substituting Eq. (10.39) into Euler's equations of motion, Eq. (9.48), and making use of the fact that the products of inertia are all equal to zero and that $I_{pu} = I_{pv}$, we obtain

$$\sum_{i=1}^{6} b_{iv}(a_{31}\,{}^i f_{bix} + a_{32}\,{}^i f_{biy} + a_{33}\,{}^i f_{biz}) = I_{pu}\dot{\omega}_{pu} - \omega_{pv}\omega_{pw}(I_{pv} - I_{pw}), \tag{10.41}$$

$$\sum_{i=1}^{6} -b_{iu}(a_{31}\,{}^i f_{bix} + a_{32}\,{}^i f_{biy} + a_{33}\,{}^i f_{biz}) = I_{pv}\dot{\omega}_{pv} - \omega_{pw}\omega_{pu}(I_{pw} - I_{pu}), \tag{10.42}$$

$$\sum_{i=1}^{6} \big[b_{iu}(a_{21}\,{}^i f_{bix} + a_{22}\,{}^i f_{biy} + a_{23}\,{}^i f_{biz}) - b_{iv}(a_{11}\,{}^i f_{bix} + a_{12}\,{}^i f_{biy} + a_{13}\,{}^i f_{biz})\big] = I_{pw}\dot{\omega}_{pw}, \tag{10.43}$$

where a_{ij} are the (i, j) elements of ${}^B R_i$, ${}^B\boldsymbol{\omega}_p = [\omega_{pu}, \omega_{pv}, \omega_{pw}]^{\mathrm{T}}$ is the angular velocity of the moving platform expressed in the moving frame B, and I_{pu}, I_{pv}, and I_{pw} are the u, v, and w components of the principal moments of inertia of the moving platform about its center of mass and expressed in the moving frame B.

Hence once ${}^{i}f_{bix}$ and ${}^{i}f_{biy}$ are determined from Eqs. (10.32) and (10.33), Eqs. (10.36) through (10.38) and (10.41) through (10.43) constitute a set of six linear equations in six unknowns, ${}^{i}f_{biz}$ for $i = 1, 2, \ldots, 6$. These six equations can easily be solved by, for example, the Gauss elimination method.

10.2.4 Actuator and Ground Reaction Forces

Once the reaction forces at the moving ball joints are found, the actuating force τ_i is obtained by summing all the forces acting on the ith piston along the z_i-axis.

$$\tau_i = {}^{i}f_{biz} + m_2 g_c c\theta_i + m_2 \, {}^{i}\dot{v}_{2iz}. \tag{10.44}$$

Note that the moment in Eq. (10.31) is taken about the fixed ball point A_i such that the reaction forces $\mathbf{f}_{ai}$ at the fixed ball joints do not enter into the equations of motion. These reaction forces, if interested, can be found by formulating Newton's equation, one limb at a time, as follows:

$${}^{i}\mathbf{f}_{ai} + {}^{i}\mathbf{f}_{bi} + (m_1 + m_2)\, {}^{i}R_A \, {}^{A}\mathbf{g} = m_1 \, {}^{i}\dot{\mathbf{v}}_{1i} + m_2 \, {}^{i}\dot{\mathbf{v}}_{2i}. \tag{10.45}$$

10.2.5 Newton–Euler's Procedure

The procedure for solving the inverse dynamics of a Stewart–Gough platform can be summarized in five basic steps:

1. Perform the inverse kinematic analysis of the mechanism. We first compute the position, velocity, and acceleration of the moving ball points in terms of a prescribed motion of the moving platform. Then we derive the position, velocity, and acceleration of the center of mass and the angular velocity and angular acceleration of each limb. Specifically, for $i = 1$ to 6, we compute:
   (a) $\mathbf{b}_i = {}^{A}R_B \, {}^{B}\mathbf{b}_i$.
   (b) d_i and $\mathbf{s}_i$ from Eqs. (10.7) and (10.6).
   (c) $c\phi_i$, $s\phi_i$, $c\theta_i$, $s\theta_i$, and ${}^{A}R_i$ from Eqs. (10.11) and (10.8).
   (d) $\mathbf{v}_{bi}$ and ${}^{i}\mathbf{v}_{bi}$ from Eqs. (10.14) and (10.15).
   (e) ${}^{i}\boldsymbol{\omega}_i$ from Eq. (10.18).
   (f) $\dot{\mathbf{v}}_{bi}$ and ${}^{i}\dot{\mathbf{v}}_{bi}$ from Eqs. (10.21) and (10.22).
   (g) ${}^{i}\dot{\boldsymbol{\omega}}_i$ from Eq. (10.25).
   (h) ${}^{i}\dot{\mathbf{v}}_{1i}$ and ${}^{i}\dot{\mathbf{v}}_{2i}$ from Eqs. (10.26) and (10.27).
2. Decompose the manipulator into a moving platform and several open-loop chains by cutting open at the moving ball joints. Assign appropri-

ate action and reaction forces at the points of connection between the moving platform and the six limbs.

3. Consider each limb as a subsystem and formulate Euler's equations of motion about the fixed ball joint for each limb. In this way, some of the reaction forces at the moving ball points can be determined independent of the equations of motion of the moving platform. Specifically, we solve Eqs. (10.32) and (10.33) for ${}^i f_{bix}$ and ${}^i f_{biy}$.
4. Solve the remaining reaction forces by formulating Newton's and Euler's equations of motion of the moving platform. That is, solve Eqs. (10.36) through (10.38) and Eqs. (10.41) through (10.43) for ${}^i f_{biz}$ for $i = 1, 2, \dots, 6$.
5. Find the actuator force τ_i by Eq. (10.44).

10.3 PRINCIPLE OF VIRTUAL WORK

In this section we illustrate how to apply the *principle of virtual work* or *d'Alembert's principle* for the dynamic analysis of parallel manipulators. The following conventions are employed:

$\mathbf{f}_i$: resulting force (excluding the actuator force) exerted at the center of mass of link i.

$\mathbf{f}_i^*$: inertia force exerted at the center of mass of link i, $\mathbf{f}_i^* = -m_i \dot{\mathbf{v}}_i$.

$\hat{\mathbf{f}}_i$: $\mathbf{f}_i + \mathbf{f}_i^*$.

$\mathbf{f}_p$: resulting force exerted at the center of mass of the moving platform.

$\mathbf{f}_p^*$: inertia force exerted at the center of mass of the moving platform, $\mathbf{f}_p^* = -m_p \dot{\mathbf{v}}_p$.

$\hat{\mathbf{f}}_p$: $\mathbf{f}_p + \mathbf{f}_p^*$.

$\mathbf{n}_i$: resulting moment (excluding the actuator torque) exerted at the center of mass of link i.

$\mathbf{n}_i^*$: inertia moment exerted at the center of mass of link i, $\mathbf{n}_i^* = -{}^iI_i\, {}^i\dot{\omega}_i - {}^i\omega_i \times ({}^iI_i\, {}^i\omega_i)$.

$\hat{\mathbf{n}}_i$: $\mathbf{n}_i + \mathbf{n}_i^*$.

$\mathbf{n}_p$: resulting moment exerted at the center of mass of the moving platform.

$\mathbf{n}_p^*$: inertia moment exerted at the center of mass of the moving platform, $\mathbf{n}_p^* = -I_p \dot{\omega}_p - \omega_p \times (I_p \omega_p)$.

$\hat{\mathbf{n}}_p$: $\mathbf{n}_p + \mathbf{n}_p^*$.

$\mathbf{q}$: vector of actuated limb lengths, $\mathbf{q} = [d_1, d_2, \dots, d_n]^{\mathrm{T}}$.

$\mathbf{x}_i$: six-dimensional vector describing the position and orientation of link i.

$\dot{\mathbf{x}}_{ji}$: six-dimensional vector describing the linear and angular velocities of the cylinder ($j = 1$) or piston ($j = 2$) of the ith limb, $\dot{\mathbf{x}}_{ji} = [v_{jix}, v_{jiy}, \ldots, \omega_{iz}]^{\mathrm{T}}$.

$\mathbf{x}_p$: six-dimensional vector describing the position and orientation of the moving platform.

$\dot{\mathbf{x}}_p$: six-dimensional vector describing the linear and angular velocities of the moving platform, $\dot{\mathbf{x}}_p = [v_{px}, v_{py}, \ldots, \omega_{pz}]^{\mathrm{T}}$.

$\boldsymbol{\tau}$: vector of actuated joint torques and/or forces.

$\delta(\cdot)$: virtual displacement of $(\cdot)$.

For convenience, we introduce a six-dimensional wrench, $\hat{\mathbf{F}}_i$, as the sum of applied and inertia wrenches about the center of mass of link i:

$$\hat{\mathbf{F}}_i = \begin{bmatrix} \hat{\mathbf{f}}_i \\ \hat{\mathbf{n}}_i \end{bmatrix}. \tag{10.46}$$

Similarly, we introduce a six-dimensional wrench, $\hat{\mathbf{F}}_p$, as the sum of applied and inertia wrenches about the center of mass of the moving platform:

$$\hat{\mathbf{F}}_p = \begin{bmatrix} \hat{\mathbf{f}}_p \\ \hat{\mathbf{n}}_p \end{bmatrix}. \tag{10.47}$$

Then the principle of virtual work for a parallel manipulator can be stated as

$$\delta\mathbf{q}^{\mathrm{T}}\boldsymbol{\tau} + \delta\mathbf{x}_p^{\mathrm{T}}\,\hat{\mathbf{F}}_p + \sum_i \delta\mathbf{x}_i^{\mathrm{T}}\,\hat{\mathbf{F}}_i = 0, \tag{10.48}$$

where the summation goes over all links of the limbs. Note that in Eq. (10.48) we have isolated the actuator torques and/or forces from other applied forces for convenience of derivation.

The virtual displacements in Eq. (10.48), however, must be compatible with the kinematic constraints imposed by the joints. Therefore, it is necessary to relate these virtual displacements to a set of independent generalized virtual displacements. For parallel manipulators, the coordinates of the moving platform, $\mathbf{x}_p = [x_p, y_p, \ldots, \psi]^{\mathrm{T}}$, can conveniently be chosen as the generalized coordinates. This is because the virtual displacement of actuated joints, $\delta\mathbf{q}$, is related to the virtual displacement of the moving platform, $\delta\mathbf{x}_p$, by the manipulator Jacobian matrix J_p:

$$\delta\mathbf{q} = J_p\,\delta\mathbf{x}_p. \tag{10.49}$$

Furthermore, the virtual displacement of the ith link of a limb, $\delta\mathbf{x}_i$, can be related to the virtual displacement of the moving platform, $\delta\mathbf{x}_p$, by a link Jacobian matrix J_i:

$$\delta\mathbf{x}_i = J_i\,\delta\mathbf{x}_p. \tag{10.50}$$

Substituting Eqs. (10.49) and (10.50) into (10.48), we obtain

$$\delta\mathbf{x}_p^{\mathrm{T}}(J_p^{\mathrm{T}}\boldsymbol{\tau} + \hat{\mathbf{F}}_p + \sum_i J_i^{\mathrm{T}}\,\hat{\mathbf{F}}_i) = 0. \tag{10.51}$$

Since Eq. (10.51) is valid for any virtual displacement $\delta\mathbf{x}_p$, it follows that

$$J_p^{\mathrm{T}}\boldsymbol{\tau} + \hat{\mathbf{F}}_p + \sum_i J_i^{\mathrm{T}}\,\hat{\mathbf{F}}_i = 0. \tag{10.52}$$

Equation (10.52) describes the dynamics of a parallel manipulator that is similar to Kane's formulation (Kane and Levinson, 1985). Note that the wrenches in Eq. (10.52) are taken about the center of mass of each link. Hence if an external wrench is applied at a point other than the center of mass, it should be transformed into the center-of-mass coordinate frame before it is substituted into the equations of motion.

In general, if the number of actuators is equal to the number of degrees of freedom of a manipulator, J_p is a square matrix. Hence $\boldsymbol{\tau}$ can be uniquely determined:

$$\boldsymbol{\tau} = -J_p^{-\mathrm{T}}(\hat{\mathbf{F}}_p + \sum_i J_i^{\mathrm{T}}\,\hat{\mathbf{F}}_i), \qquad \text{provided that } J_p \text{ is nonsingular.} \tag{10.53}$$

On the other hand, if the number of actuators is greater than the number of degrees of freedom, $\boldsymbol{\tau}$ has an infinitude of solutions from which a minimum norm solution can be obtained by applying the pseudoinverse technique.

In what follows, we use the Stewart–Gough platform shown in Fig. 10.2 as an example to illustrate the methodology. Since the inverse kinematic analysis of the manipulator was carried out in the preceding section, we proceed directly to the formulation of link Jacobian matrices, link inertia wrenches, and the equations of motion.

10.3.1 Link Jacobian Matrices

A critical step in formulating d'Alembert's equations of motion is the derivation of the manipulator Jacobian matrix and link Jacobian matrices. To identify the manipulator Jacobian matrix, J_p, we write Eq. (10.14) in matrix form:

$$\mathbf{v}_{bi} = J_{bi}\,\dot{\mathbf{x}}_p, \tag{10.54}$$

where

$$J_{bi} = \begin{bmatrix} 1 & 0 & 0 & 0 & b_{iz} & -b_{iy} \\ 0 & 1 & 0 & -b_{iz} & 0 & b_{ix} \\ 0 & 0 & 1 & b_{iy} & -b_{ix} & 0 \end{bmatrix}. \tag{10.55}$$

Substituting Eq. (10.54) into (10.15), we obtain

$${}^i\mathbf{v}_{bi} = {}^iJ_{bi}\,\dot{\mathbf{x}}_p = \begin{bmatrix} {}^iJ_{bix} \\ {}^iJ_{biy} \\ {}^iJ_{biz} \end{bmatrix} \dot{\mathbf{x}}_p, \tag{10.56}$$

where ${}^iJ_{bi} = {}^iR_A J_{bi}$ is a 3×6 matrix whose row vectors are given by

$$\begin{aligned} {}^iJ_{bix} &= [\mathrm{c}\phi_i\mathrm{c}\theta_i, \mathrm{s}\phi_i\mathrm{c}\theta_i, -\mathrm{s}\theta_i, -b_{iz}\mathrm{s}\phi_i\mathrm{c}\theta_i - b_{iy}\mathrm{s}\theta_i, b_{iz}\mathrm{c}\phi_i\mathrm{c}\theta_i + b_{ix}\mathrm{s}\theta_i, \\ &\quad - b_{iy}\mathrm{c}\phi_i\mathrm{c}\theta_i + b_{ix}\mathrm{s}\phi_i\mathrm{c}\theta_i], \\ {}^iJ_{biy} &= [-\mathrm{s}\phi_i, \mathrm{c}\phi, 0, -b_{iz}\mathrm{c}\phi_i, -b_{iz}\mathrm{s}\phi_i, b_{iy}\mathrm{s}\phi_i + b_{ix}\mathrm{c}\phi], \\ {}^iJ_{biz} &= [\mathrm{c}\phi_i\mathrm{s}\theta_i, \mathrm{s}\phi_i\mathrm{s}\theta_i, \mathrm{c}\theta_i, -b_{iz}\mathrm{s}\phi_i\mathrm{s}\theta_i + b_{iy}\mathrm{c}\theta_i, b_{iz}\mathrm{c}\phi_i\mathrm{s}\theta_i - b_{ix}\mathrm{c}\theta_i, \\ &\quad - b_{iy}\mathrm{c}\phi_i\mathrm{s}\theta_i + b_{ix}\mathrm{s}\phi_i\mathrm{s}\theta_i]. \end{aligned} \tag{10.57}$$

Making use of the row matrices above, Eq. (10.17) can be written as

$$\dot{d}_i = {}^iJ_{biz}\dot{\mathbf{x}}_p. \tag{10.58}$$

Writing Eq. (10.58) six times, once for each limb, yields six scalar equations which can be assembled in the matrix form

$$\dot{\mathbf{q}} = J_p\dot{\mathbf{x}}_p, \tag{10.59}$$

where

$$J_p = \begin{bmatrix} {}^1J_{b1z} \\ {}^2J_{b2z} \\ \vdots \\ {}^6J_{b6z} \end{bmatrix} \tag{10.60}$$

is known as the *manipulator Jacobian matrix*.

Similarly, Eqs. (10.18), (10.19), and (10.20) can be written as

$${}^i\boldsymbol{\omega}_i = \frac{1}{d_i} \begin{bmatrix} -{}^iJ_{biy} \\ {}^iJ_{bix} \\ \mathbf{0}_{1\times 6} \end{bmatrix} \dot{\mathbf{x}}_p, \tag{10.61}$$

$$
{}^i\mathbf{v}_{1i} = \frac{e_1}{d_i}\begin{bmatrix} {}^iJ_{bix} \\ {}^iJ_{biy} \\ \mathbf{0}_{1\times 6} \end{bmatrix} \dot{\mathbf{x}}_p, \tag{10.62}
$$

$$
{}^i\mathbf{v}_{2i} = \frac{1}{d_i}\begin{bmatrix} (d_i - e_2)\,{}^iJ_{bix} \\ (d_i - e_2)\,{}^iJ_{biy} \\ d_i\,{}^iJ_{biz} \end{bmatrix} \dot{\mathbf{x}}_p. \tag{10.63}
$$

Combining Eqs. (10.61), (10.62), and (10.63), we obtain

$$
{}^i\dot{\mathbf{x}}_{1i} = {}^iJ_{1i}\,\dot{\mathbf{x}}_p, \tag{10.64}
$$

$$
{}^i\dot{\mathbf{x}}_{2i} = {}^iJ_{2i}\,\dot{\mathbf{x}}_p, \tag{10.65}
$$

where

$$
{}^iJ_{1i} = \frac{1}{d_i}\begin{bmatrix} e_1\,{}^iJ_{bix} \\ e_1\,{}^iJ_{biy} \\ \mathbf{0}_{1\times 6} \\ -{}^iJ_{biy} \\ {}^iJ_{bix} \\ \mathbf{0}_{1\times 6} \end{bmatrix}, \tag{10.66}
$$

$$
{}^iJ_{2i} = \frac{1}{d_i}\begin{bmatrix} (d_i - e_2)\,{}^iJ_{bix} \\ (d_i - e_2)\,{}^iJ_{biy} \\ d_i\,{}^iJ_{biz} \\ -{}^iJ_{biy} \\ {}^iJ_{bix} \\ \mathbf{0}_{1\times 6} \end{bmatrix} \tag{10.67}
$$

are called the *link Jacobian matrices* of the ith cylinder and piston, respectively.

10.3.2 Inertia and Applied Wrenches

Assume that gravitational force is the only force acting on the manipulator. Then the vector sum of applied and inertia wrenches exerted at the center of mass of the moving platform is given by

$$
\hat{\mathbf{F}}_p = \begin{bmatrix} \hat{\mathbf{f}}_p \\ \hat{\mathbf{n}}_p \end{bmatrix} = \begin{bmatrix} m_p\mathbf{g} - m_p\dot{\mathbf{v}}_p \\ -{}^AI_p\dot{\boldsymbol{\omega}}_p - \boldsymbol{\omega}_p \times ({}^AI_p\boldsymbol{\omega}_p) \end{bmatrix}, \tag{10.68}
$$

where ${}^AI_p = {}^AR_B\,{}^BI_p\,{}^BR_A$ denotes the inertia matrix of the moving platform taken about its center of mass and expressed in the fixed frame A.

Similarly, the vector sum of external and inertia wrenches exerted at the centers of mass of the cylinder and piston and expressed in the ith limb frame are given by

$$ {}^i\hat{\mathbf{F}}_{1i} = \begin{bmatrix} {}^i\hat{\mathbf{f}}_{1i} \\ {}^i\hat{\mathbf{n}}_{1i} \end{bmatrix} = \begin{bmatrix} m_{1i}\,{}^iR_A\mathbf{g} - m_{1i}\,{}^i\dot{\mathbf{v}}_{1i} \\ -{}^iI_{1i}\,{}^i\dot{\boldsymbol{\omega}}_i - {}^i\boldsymbol{\omega}_i \times ({}^iI_{1i}\,{}^i\boldsymbol{\omega}_i) \end{bmatrix}, \tag{10.69} $$

$$ {}^i\hat{\mathbf{F}}_{2i} = \begin{bmatrix} {}^i\hat{\mathbf{f}}_{2i} \\ {}^i\hat{\mathbf{n}}_{2i} \end{bmatrix} = \begin{bmatrix} m_{2i}\,{}^iR_A\mathbf{g} - m_{2i}\,{}^i\dot{\mathbf{v}}_{2i} \\ -{}^iI_{2i}\,{}^i\dot{\boldsymbol{\omega}}_i - {}^i\boldsymbol{\omega}_i \times ({}^iI_{2i}\,{}^i\boldsymbol{\omega}_i) \end{bmatrix}. \tag{10.70} $$

Note that due to symmetry, the inertia matrices of the moving platform and the six limbs are all diagonal.

10.3.3 Equations of Motion

We now apply the principal of virtual work for derivation of the equations of motion. Equation (10.52) can be written specifically for the Stewart–Gough platform as follows:

$$ J_p^{\mathrm{T}}\boldsymbol{\tau} + \hat{\mathbf{F}}_p + \sum_{i=1}^{6}({}^iJ_{1i}^{\mathrm{T}}\,{}^i\hat{\mathbf{F}}_{1i} + {}^iJ_{2i}^{\mathrm{T}}\,{}^i\hat{\mathbf{F}}_{2i}) = 0. \tag{10.71} $$

Note that in Eq. (10.71), $\hat{\mathbf{F}}_p$ is expressed in the fixed frame A, while ${}^i\hat{\mathbf{F}}_{1i}$ and ${}^i\hat{\mathbf{F}}_{2i}$ are expressed in the limb frame i and then transformed into the fixed frame by the transpose of the link Jacobian matrices. Similarly, the vector of actuator forces, $\boldsymbol{\tau}$, is specified in the actuator joint space and then transformed into the fixed frame by the transpose of the manipulator Jacobian matrix.

Substituting Eqs. (10.66) and (10.67) into (10.71) and simplifying yields

$$ J_p^{\mathrm{T}}(\boldsymbol{\tau} + \hat{\mathbf{F}}_z) + \hat{\mathbf{F}}_p + J_x^{\mathrm{T}}\hat{\mathbf{F}}_x + J_y^{\mathrm{T}}\hat{\mathbf{F}}_y = 0, \tag{10.72} $$

where

$$ J_x = \begin{bmatrix} {}^1J_{b1x} \\ {}^2J_{b2x} \\ \vdots \\ {}^6J_{b6x} \end{bmatrix}, \tag{10.73} $$

$$ J_y = \begin{bmatrix} {}^1J_{b1y} \\ {}^2J_{b2y} \\ \vdots \\ {}^6J_{b6y} \end{bmatrix}, \tag{10.74} $$

$$\hat{\mathbf{F}}_x = \begin{bmatrix} \dfrac{e_1 \hat{f}_{11x} + (d_1 - e_2)\hat{f}_{21x} + \hat{n}_{11y} + \hat{n}_{21y}}{d_1} \\ \dfrac{e_1 \hat{f}_{12x} + (d_2 - e_2)\hat{f}_{22x} + \hat{n}_{12y} + \hat{n}_{22y}}{d_2} \\ \vdots \\ \dfrac{e_1 \hat{f}_{16x} + (d_6 - e_2)\hat{f}_{26x} + \hat{n}_{16y} + \hat{n}_{26y}}{d_6} \end{bmatrix}, \tag{10.75}$$

$$\hat{\mathbf{F}}_y = \begin{bmatrix} \dfrac{e_1 \hat{f}_{11y} + (d_1 - e_2)\hat{f}_{21y} - \hat{n}_{11x} - \hat{n}_{21x}}{d_1} \\ \dfrac{e_1 \hat{f}_{12y} + (d_2 - e_2)\hat{f}_{22y} - \hat{n}_{12x} - \hat{n}_{22x}}{d_2} \\ \vdots \\ \dfrac{e_1 \hat{f}_{16y} + (d_6 - e_2)\hat{f}_{26y} - \hat{n}_{16x} - \hat{n}_{26x}}{d_6} \end{bmatrix}, \tag{10.76}$$

$$\hat{\mathbf{F}}_z = \begin{bmatrix} \hat{f}_{21z} \\ \hat{f}_{22z} \\ \vdots \\ \hat{f}_{26z} \end{bmatrix}, \tag{10.77}$$

where $\hat{f}_{jix}$, $\hat{f}_{jiy}$, and $\hat{f}_{jiz}$ are the x-, y-, and z-components of ${}^{i}\hat{\mathbf{f}}_{ji}$, and $\hat{n}_{jix}$, $\hat{n}_{jiy}$, and $\hat{n}_{jiz}$ are the x-, y-, and z-components of ${}^{i}\hat{\mathbf{n}}_{ji}$. From Eq. (10.72), the actuator forces required to produce a desired motion characteristics can be computed by using, for example, the Gauss elimination method. Note that the actuator forces depend on the inverse of the transpose of the manipulator Jacobian matrix, $J_p^{-\mathrm{T}}$. Therefore, the computation of actuator forces may become numerically unstable when the manipulator approaches a singular configuration.

We observe that using the principle of virtual work, the constraint forces and moments are eliminated from the equations of motion. This method is, therefore, more efficient than the Newton–Euler formulation and the Lagrangian formulation. It is potentially useful for real-time control of a parallel manipulator.

10.3.4 d'Alembert's Procedure

In this section we summarize the procedure for solving the inverse dynamics of a Stewart–Gough platform by using the principle of virtual work. It is assumed that the time history of the moving platform is specified in terms of the position of the centroid, $\mathbf{p}$, and three Euler's angles, ϕ, θ, and ψ. The velocity and acceleration of the centroid are calculated by taking the derivatives of $\mathbf{p}$ with respect to time. The rotation matrix of the moving platform, ${}^AR_B$, is calculated from Eq. (10.1). The angular velocity and angular acceleration of the moving platform are calculated from Eqs. (10.3) and (10.4). At any instant in time, the actuator forces are computed by the following five steps:

1. Determine the position, velocity, and acceleration of all links by performing the inverse kinematic analysis. Specifically, for $i = 1$ to 6, we compute:
   (a) $\mathbf{b}_i = {}^AR_B\,{}^B\mathbf{b}_i$ and ${}^AI_p = {}^AR_B\,{}^BI_p\,{}^BR_A$.
   (b) d_i and $\mathbf{s}_i$ from Eqs. (10.7) and (10.6).
   (c) $c\phi_i$, $s\phi_i$, $c\theta_i$, $s\theta_i$, and ${}^AR_i$ from Eqs. (10.11) and (10.8).
   (d) $\mathbf{v}_{bi}$ and ${}^i\mathbf{v}_{bi}$ from Eqs. (10.14) and (10.15).
   (e) ${}^i\boldsymbol{\omega}_i$ from Eq. (10.18).
   (f) $\dot{\mathbf{v}}_{bi}$ and ${}^i\dot{\mathbf{v}}_{bi}$ from Eqs. (10.21) and (10.22).
   (g) ${}^i\dot{\boldsymbol{\omega}}_i$ from Eq. (10.25).
   (h) ${}^i\dot{\mathbf{v}}_{1i}$ and ${}^i\dot{\mathbf{v}}_{2i}$ from Eqs. (10.26) and (10.27).
2. Determine the platform and link Jacobian matrices.
   (a) For $i = 1$ to 6, calculate J_{bi} and then ${}^iJ_{bi}$ from Eqs. (10.55) and (10.57).
   (b) Calculate J_p by Eq. (10.60).
3. Determine the resultants of the applied and inertia wrenches.
   (a) Calculate $\hat{\mathbf{F}}_p$ by Eq. (10.68).
   (b) For $i = 1$ to 6, calculate ${}^i\hat{\mathbf{F}}_{1i}$ and ${}^i\hat{\mathbf{F}}_{2i}$ from Eqs. (10.69) and (10.70), respectively.
4. Formulate the dynamical equations of motion. Calculate J_x, J_y, $\hat{\mathbf{F}}_x$, $\hat{\mathbf{F}}_y$, and $\hat{\mathbf{F}}_z$ from Eqs. (10.73) through (10.77), and then substitute the resulting expressions into Eq. (10.72).
5. Solve the resulting dynamical equations of motion, Eq. (10.72), by the Gaussian elimination method.

10.3.5 Numerical Examples

Based on the algorithm above, a computer program was developed to solve the inverse dynamics of the Stewart–Gough platform shown in Fig. 10.2, using MATLAB software. Figure 10.5 shows the locations of the ball joints and the initial location of the moving platform. The manipulator system parameters are $m_p = 32$ kg, $m_{1,i} = 2$ kg, $m_{2,i} = 2$ kg, $b_i = 0.5$ m, $a_i = 1$ m, $e_1 =$

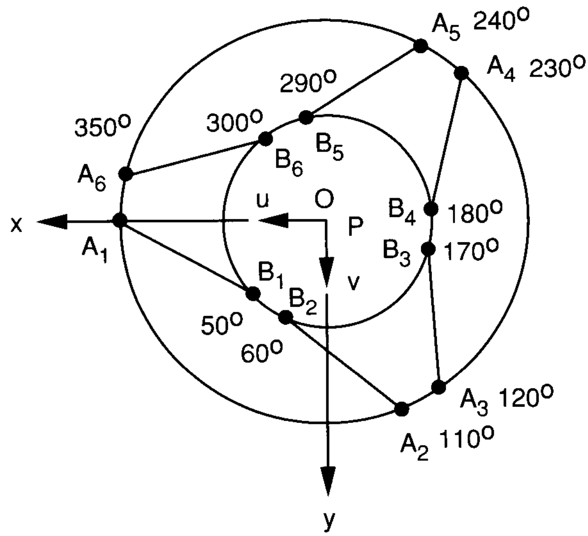

FIGURE 10.5. Top view of the Stewart–Gough platform.

0.2 m, $e_2 = 0.75$ m, ${}^B I_p = \text{diag}[2, 2, 4]\,\text{kg}\cdot\text{m}^2$, ${}^i I_{1i} = \text{diag}[1, 1, 0.001]\,\text{kg}\cdot\text{m}^2$, ${}^i I_{2i} = \text{diag}[0.4, 0.4, 0.0001]\,\text{kg}\cdot\text{m}^2$, and $\mathbf{g} = [0, 0, -9.8]^T$ m/s^2.

The following examples are solved to illustrate the algorithm. In all examples it is assumed that the platform starts at rest and accelerates with a constant acceleration for a period of 0.4 second. Furthermore, at the initial location, the moving platform is assumed to be located 1 m above the fixed base, namely, at $t = 0$, $\mathbf{p} = [0, 0, 1]^T$ m, and $\phi = \theta = \psi = 0$.

For the first example, the moving platform moves along the x-direction with a constant acceleration of $\ddot{p}_x = 5$ m/s^2 while all the other parameters are held constant, $\ddot{p}_y = \ddot{p}_z = \ddot{\phi} = \ddot{\theta} = \ddot{\psi} = 0$. The actuating forces versus time calculated by the program are plotted in Fig. 10.6.

For the second example, the moving platform moves along the z-direction with a constant acceleration of $\ddot{p}_z = 5$ m/s^2 while all the other parameters are held constant, $\ddot{p}_x = \ddot{p}_y = \ddot{\phi} = \ddot{\theta} = \ddot{\psi} = 0$. As can be seen from Fig. 10.7, to accelerate the platform along the z-axis, all actuating forces are equal to one another. As the moving platform moves away from the fixed base, the limbs become more vertically oriented, therefore reducing the actuating forces.

For the third example, the moving platform rotates about the y-axis with a constant angular acceleration of $\ddot{\theta} = 2$ rad/s^2 while all the other parameters are held constant, $\ddot{p}_x = \ddot{p}_y = \ddot{p}_z = \ddot{\phi} = \ddot{\psi} = 0$. The actuating forces versus time are plotted in Fig. 10.8.

For the fourth example, the moving platform rotates about the z-axis with a constant angular acceleration of $\ddot{\phi} = 2$ rad/s^2 while all the other parameters

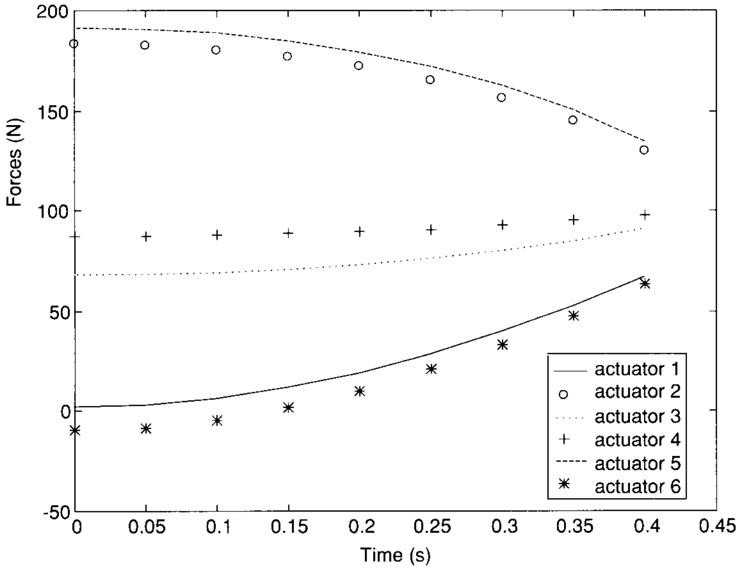

FIGURE 10.6. Actuating forces versus time, $\ddot{p}_x = 5$ m/s^2.

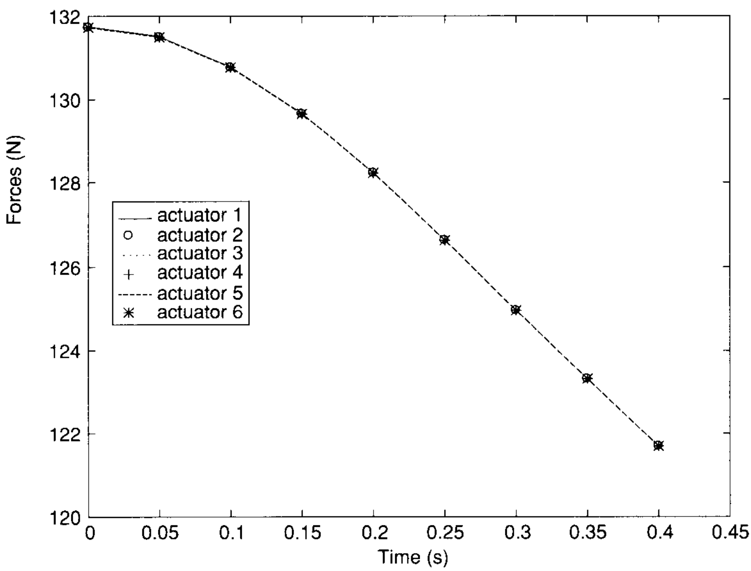

FIGURE 10.7. Actuating forces versus time, $\ddot{p}_z = 5$ m/s^2.

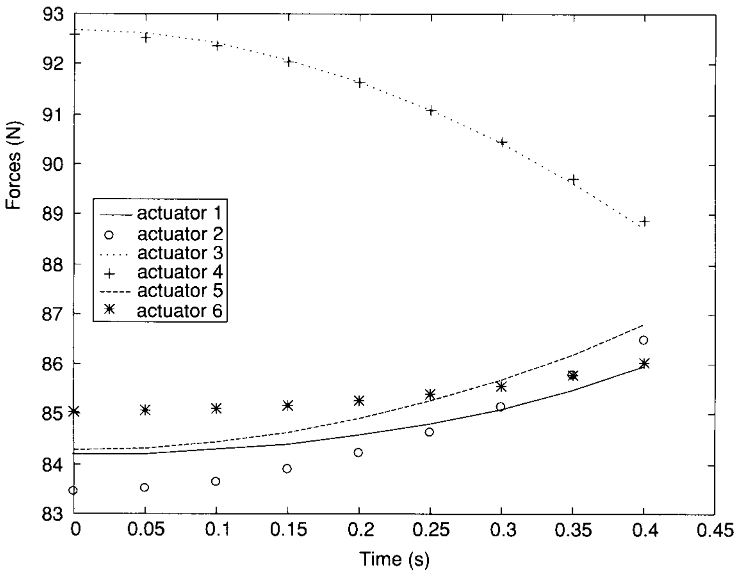

FIGURE 10.8. Actuating forces versus time, $\ddot{\theta} = 2$ rad/s^2.

are held constant, $\ddot{p}_x = \ddot{p}_y = \ddot{p}_z = \ddot{\theta} = \ddot{\psi} = 0$. As can be seen from Fig. 10.9, due to the symmetrical arrangement of the limbs, the forces exerted by actuators 1, 3, and 5 are equal to one another, and the forces exerted by actuators 2, 4, and 6 are also equal to one another.

10.4 LAGRANGIAN FORMULATION

In this section we show that the inverse dynamics of relatively simple parallel manipulators can be solved by applying the Lagrangian equations of the first type. The Lagrangian equations of the first type are written in terms of a set of redundant coordinates. Therefore, the formulation requires a set of constraint equations derived from the kinematics of a mechanism. These constraint equations and their derivatives must be adjoined to the equations of motion to produce a number of equations that is equal to the number of unknowns.

The Lagrangian equations of the first type can be written as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = Q_j + \sum_{i=1}^{k} \lambda_i \frac{\partial \Gamma_i}{\partial q_j} \qquad \text{for } j = 1 \text{ to } n, \quad (10.78)$$

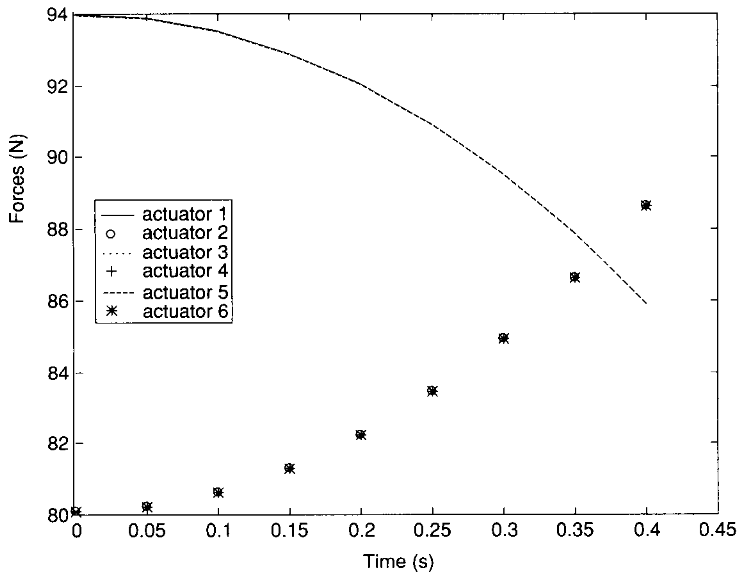

FIGURE 10.9. Actuating forces versus time, $\ddot{\phi} = 2$ rad/s^2.

where Γ_i denotes the ith constraint function, k is the number of constraint functions, and λ_i is the Lagrangian multiplier. The number of coordinates, n, exceeds the number of degrees of freedom by k. Solving the equations of motion is made easier by arranging the Lagrangian equations into two sets. One contains the Lagrange multipliers as the only unknowns, and the other contains the generalized forces contributed by the actuators as the additional unknowns. Let the first k equations be associated with the redundant coordinates and the remaining $n - k$ equations be associated with the actuated joint variables. Then the first set of equations can be written in the form

$$\sum_{i=1}^{k} \lambda_i \frac{\partial \Gamma_i}{\partial q_j} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} - \hat{Q}_j, \tag{10.79}$$

where $\hat{Q}_j$, if any, represents the generalized force contributed by an externally applied force. For the inverse dynamics, $\hat{Q}_j$ is given. Hence the right-hand side of Eq. (10.79) is known. Writing Eq. (10.79) once for each redundant coordinate yields a set of k linear equations that can be solved for the k Langrangian multipliers.

Once the Lagrange multipliers are found, the actuator torques and/or forces can be determined directly from the remaining equations. Specifically, the second set of equations can be written as

$$Q_j = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} - \sum_{i=1}^{k} \lambda_i \frac{\partial \Gamma_i}{\partial q_j} \qquad \text{for } j = k+1 \text{ to } n, \tag{10.80}$$

where Q_j is the actuator force or torque.

In what follows, we analyze the dynamics of a 3-dof parallel manipulator as an example to illustrate the principle.

10.4.1 Lagrangian Dynamics of the University of Maryland Manipulator

In this section we analyze the dynamics of the University of Maryland manipulator as an example to illustrate the methodology. For convenience, the schematic diagram is repeated in Fig. 10.10. The coordinate frames, link lengths, and the joint angles of the manipulator are shown in more detail in Fig. 3.11. In this manipulator, θ_{11}, θ_{12}, and θ_{13} are the actuated joints.

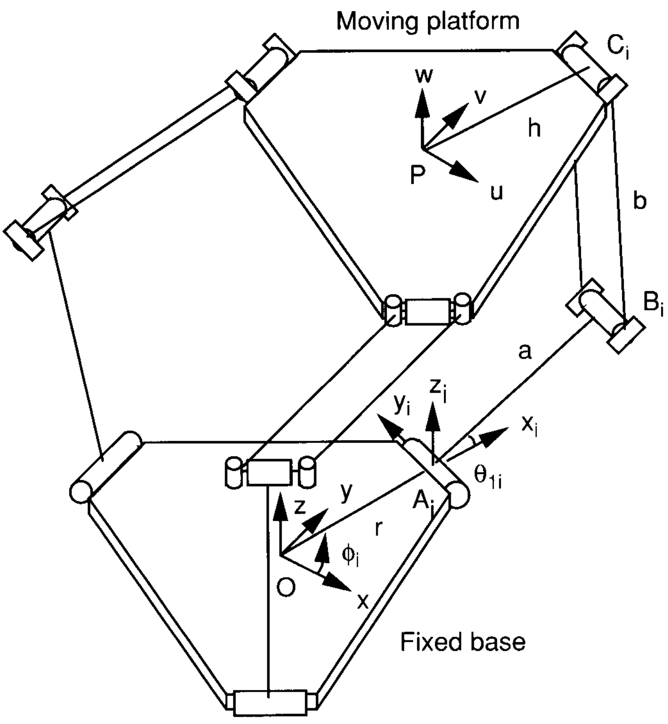

FIGURE 10.10. University of Maryland's manipulator.

Theoretically, the dynamic analysis can be accomplished by using just three generalized coordinates since this is a 3-dof manipulator. However, this would lead to a cumbersome expression for the Lagrange function, due to the complex kinematics of the manipulator. Instead, Lagrange's equations of the first type will be employed by introducing three redundant coordinates, p_x, p_y, and p_z. Thus we have $p_x, p_y, p_z, \theta_{11}, \theta_{12}$, and θ_{13} as the generalized coordinates. Equation (10.78) represents a system of six equations in six variables. The six variables are λ_i for $i = 1$ to 3, and the three actuator torques, Q_j for $j = 4, 5$, and 6. Note that the generalized forces, Q_i for $i = 1$ to 3, represent the x, y, and z components of an external force exerted at the centroid P of the moving platform.

This formulation requires three constraint equations, Γ_i for $i = 1$ to 3. The constraint equations are obtained from the fact that the distance between joints B and C is always equal to the length of the connecting rod of the upper arm, b; that is,

$$\begin{aligned}\Gamma_i &= \overline{B_i C_i}^2 - b^2 \\ &= (p_x + hc\phi_i - rc\phi_i - ac\phi_i c\theta_{1i})^2 + (p_y + hs\phi_i - rs\phi_i - as\phi_i c\theta_{1i})^2 \\ &\quad + (p_z - as\theta_{1i})^2 - b^2 \\ &= 0 \end{aligned} \tag{10.81}$$

for $i = 1, 2$, and 3.

To simplify the analysis, we assume that the mass of each connecting rod, m_b, in the upper arm assembly is divided evenly and concentrated at the two endpoints B_i and C_i. Then we derive the Lagrangian function L as follows. The total kinetic energy of the manipulator is

$$K = K_p + \sum_{i=1}^{3} (K_{ai} + K_{bi}), \tag{10.82}$$

where K_p is the kinetic energy of the moving platform, K_{ai} the kinetic energy of the input link and the rotor on limb i, and K_{bi} the kinetic energy of the two connecting rods of limb i. Specifically,

$$\begin{aligned} K_p &= \tfrac{1}{2} m_p (\dot{p}_x^2 + \dot{p}_y^2 + \dot{p}_z^2), \\ K_{ai} &= \tfrac{1}{2} (I_m + \tfrac{1}{3} m_a a^2) \dot{\theta}_{1i}^2, \\ K_{bi} &= \tfrac{1}{2} m_b (\dot{p}_x^2 + \dot{p}_y^2 + \dot{p}_z^2) + \tfrac{1}{2} m_b a^2 \dot{\theta}_{1i}^2, \end{aligned}$$

where m_p is the mass of the moving platform, m_a the mass of the input link, m_b the mass of one of the two connecting rods, and I_m the axial moment of inertia of the rotor mounted on the ith limb. Assuming that the acceleration of gravity points in the $-z$-direction, the total potential energy of the manipulator relative to the fixed x–y plane is

$$U = U_p + \sum_{i=1}^{3} (U_{ai} + U_{bi}), \tag{10.83}$$

where U_p is the potential energy of the moving platform, U_{ai} the potential energy of the input link on limb i, and U_{bi} the potential energy of the two connecting rods of the ith limb. Specifically,

$$\begin{aligned} U_p &= m_p\, g_c\, p_z, \\ U_{ai} &= \tfrac{1}{2} m_a\, g_c\, a s\theta_{1i}, \\ U_{bi} &= m_b\, g_c (p_z + a s\theta_{1i}). \end{aligned}$$

Therefore, the Lagrangian function is

$$\begin{aligned} L = {} & \tfrac{1}{2}(m_p + 3m_b)(\dot{p}_x^2 + \dot{p}_y^2 + \dot{p}_z^2) \\ & + \tfrac{1}{2}\left(I_m + \tfrac{1}{3} m_a a^2 + m_b a^2\right)(\dot{\theta}_{11}^2 + \dot{\theta}_{12}^2 + \dot{\theta}_{13}^2) - (m_p + 3m_b) g_c p_z \\ & - \left(\tfrac{1}{2} m_a + m_b\right) g_c a (s\theta_{11} + s\theta_{12} + s\theta_{13}). \end{aligned} \tag{10.84}$$

Taking the derivatives of the Lagrangian function with respect to the six generalized coordinates, we obtain

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{p}_x}\right) &= (m_p + 3m_b)\ddot{p}_x, & \frac{\partial L}{\partial p_x} &= 0, \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{p}_y}\right) &= (m_p + 3m_b)\ddot{p}_y, & \frac{\partial L}{\partial p_y} &= 0, \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{p}_z}\right) &= (m_p + 3m_b)\ddot{p}_z, & \frac{\partial L}{\partial p_z} &= -(m_p + 3m_b) g_c, \end{aligned}$$

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_{11}}\right) &= \left(I_m + \tfrac{1}{3} m_a a^2 + m_b a^2\right)\ddot{\theta}_{11}, & \frac{\partial L}{\partial \theta_{11}} &= -\left(\tfrac{1}{2} m_a + m_b\right) g_c a c\theta_{11}, \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_{12}}\right) &= \left(I_m + \tfrac{1}{3} m_a a^2 + m_b a^2\right)\ddot{\theta}_{12}, & \frac{\partial L}{\partial \theta_{12}} &= -\left(\tfrac{1}{2} m_a + m_b\right) g_c a c\theta_{12}, \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_{13}}\right) &= \left(I_m + \tfrac{1}{3} m_a a^2 + m_b a^2\right)\ddot{\theta}_{13}, & \frac{\partial L}{\partial \theta_{13}} &= -\left(\tfrac{1}{2} m_a + m_b\right) g_c a c\theta_{13}. \end{aligned}$$

Taking the partial derivatives of the constraint function Γ_i with respect to the six generalized coordinates yields

$$\frac{\partial \Gamma_i}{\partial p_x} = 2(p_x + hc\phi_i - rc\phi_i - ac\phi_i c\theta_{1i}) \qquad \text{for } i = 1, 2, 3,$$

$$\frac{\partial \Gamma_i}{\partial p_y} = 2(p_y + hs\phi_i - rs\phi_i - as\phi_i c\theta_{1i}) \qquad \text{for } i = 1, 2, 3,$$

$$\frac{\partial \Gamma_i}{\partial p_z} = 2(p_z - as\theta_{1i}) \qquad \text{for } i = 1, 2, 3,$$

$$\frac{\partial \Gamma_1}{\partial \theta_{11}} = 2a\left[(p_x c\phi_1 + p_y s\phi_1 + h - r)s\theta_{11} - p_z c\theta_{11}\right],$$

$$\frac{\partial \Gamma_i}{\partial \theta_{11}} = 0 \qquad \text{for } i = 2, 3,$$

$$\frac{\partial \Gamma_i}{\partial \theta_{12}} = 0 \qquad \text{for } i = 1, 3,$$

$$\frac{\partial \Gamma_2}{\partial \theta_{12}} = 2a\left[(p_x c\phi_2 + p_y s\phi_2 + h - r)s\theta_{12} - p_z c\theta_{12}\right],$$

$$\frac{\partial \Gamma_i}{\partial \theta_{13}} = 0 \qquad \text{for } i = 1, 2,$$

$$\frac{\partial \Gamma_3}{\partial \theta_{13}} = 2a\left[(p_x c\phi_3 + p_y s\phi_3 + h - r)s\theta_{13} - p_z c\theta_{13}\right].$$

Substituting the derivatives above into Eqs. (10.79) and (10.80), we obtain a system of dynamical equations. For $j = 1, 2$, and 3:

$$2\sum_{i=1}^{3} \lambda_i (p_x + hc\phi_i - rc\phi_i - ac\phi_i c\theta_{1i}) = (m_p + 3m_b)\ddot{p}_x - f_{px}, \tag{10.85}$$

$$2\sum_{i=1}^{3} \lambda_i (p_y + hs\phi_i - rs\phi_i - as\phi_i c\theta_{1i}) = (m_p + 3m_b)\ddot{p}_y - f_{py}, \tag{10.86}$$

$$2\sum_{i=1}^{3} \lambda_i (p_z - as\theta_{1i}) = (m_p + 3m_b)\ddot{p}_z + (m_p + 3m_b)g_c - f_{pz}, \tag{10.87}$$

where f_{px} f_{py}, and f_{pz} are the x-, y-, and z-components of an external force exerted on the moving platform. For $j = 4, 5$, and 6:

$$\tau_1 = \left(I_m + \tfrac{1}{3}m_a a^2 + m_b a^2\right)\ddot{\theta}_{11} + \left(\tfrac{1}{2}m_a + m_b\right) g_c a c\theta_{11} \\ - 2a\lambda_1\left[(p_x c\phi_1 + p_y s\phi_1 + h - r)s\theta_{11} - p_z c\theta_{11}\right], \quad (10.88)$$

$$\tau_2 = \left(I_m + \tfrac{1}{3}m_a a^2 + m_b a^2\right)\ddot{\theta}_{12} + \left(\tfrac{1}{2}m_a + m_b\right) g_c a c\theta_{12} \\ - 2a\lambda_2\left[(p_x c\phi_2 + p_y s\phi_2 + h - r)s\theta_{12} - p_z c\theta_{12}\right], \quad (10.89)$$

$$\tau_3 = \left(I_m + \tfrac{1}{3}m_a a^2 + m_b a^2\right)\ddot{\theta}_{13} + \left(\tfrac{1}{2}m_a + m_b\right) g_c a c\theta_{13} \\ - 2a\lambda_3\left[(p_x c\phi_3 + p_y s\phi_3 + h - r)s\theta_{13} - p_z c\theta_{13}\right]. \quad (10.90)$$

Equations (10.85) through (10.87) form a set of three linear equations in three unknowns from which the three Lagrange multipliers can be determined. Once the Lagrange multipliers are found, the actuator torques are determined from the second set of equations, Eqs. (10.88) through (10.90). These two sets of equations can be used for real-time control of the manipulator (Stamper, 1997).

10.5 SUMMARY

The inverse dynamics of parallel manipulators has been studied in this chapter. First, a numerical solution technique based on the Newton–Euler equations was presented. It was shown that by considering each limb as a subsystem, the reaction forces and moments at the joint connecting the upper and lower members of a limb can be eliminated from the equations of motion. It was also shown that by writing the equations of motion of each limb about the fixed ball joint and by expressing the resulting equations in the link frame, reaction forces at the fixed ball joint are eliminated at the outset while some of the reaction forces at the moving ball joint can be solved independently of the equations of motion of the moving platform. Hence the computational efficiency is greatly improved.

Second, a more efficient method based on the principle of virtual work has been developed. A critical step in applying this method is the development of link Jacobian matrices that relate the velocity states of the limbs to the velocity state of the moving platform. A Stewart–Gough platform with six extensible limbs was used as an example to illustrate the methodology. It was shown that using the principle of virtual work, dynamical equations of motion for such a complex manipulator can be reduced to solving a system of six equations in six unknowns. To demonstrate the algorithm, a computer algorithm was developed and several numerical examples were solved. It is believed that this method is more efficient than the Newton–Euler method. It can potentially be employed for real-time control of a Stewart–Gough platform.

Although it is nearly impossible to derive explicit dynamical equations for a general 6-dof parallel manipulator, it has been shown that explicit equations of motion for some relatively simple manipulators can be derived by applying Lagrangian equations of the first type. The dynamics of the University of Maryland manipulator were analyzed to illustrate the method.

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EXERCISES

1. Derive the dynamic equations of motion for the planar five-bar manipulator shown in Fig. 1.7 by the Newton–Euler formulation. To simplify the analysis, let the inertias of links 1, 2, and 3 be negligible, link 4 be a slender homogeneous rod of mass m, and the acceleration of gravity be pointing in the negative y-direction.

2. For the planar 3*RPR* manipulator shown in Fig. 6.12, let the inertias of the limbs be negligible, the moving platform be a triangular plate of uniform

density of mass m, and the acceleration of gravity be pointing out of the plane of motion. Develop the dynamical equations of motion by the Newton–Euler formulation.

3. Derive the dynamical equations of motion for the planar five-bar manipulator shown in Fig. 3.19 by the principle of virtual work. To simplify the analysis, let the inertias of links 1, 2, and 3 be negligible, link 4 be a slender homogeneous rod of mass m, and the acceleration of gravity be pointing in the $-y$-direction.

4. Derive the dynamical equations of motion for the University of Maryland manipulator shown in Fig. 3.10 using the principle of virtual work. Assume that the inertias of the limbs are negligible, the moving platform is triangular plate of uniform density of mass m, and the acceleration of gravity points in the $-z$-direction.

5. For the planar five-bar manipulator shown in Fig. 3.19, let a point mass m be attached to the end effector at point Q, the inertias of all links be negligible, and the acceleration of gravity be pointing in the $-y$-direction. Derive the dynamical equations of motion by the Lagrangian formulation.

APPENDIX A

CONTINUATION METHOD

The inverse kinematics problem of a serial manipulator can often be reduced to a single polynomial in one variable if the manipulator has three consecutive joint axes intersecting at a common point or parallel to one another. However, for a manipulator of general kinematic structure, closed-form solutions become a nearly impossible task. In this case, it may be necessary to solve the problem numerically. In this appendix we introduce a numerical method known as the *continuation* or *homotopy method*. Readers are referred to Garcia and Li (1980), Garcia and Zangwill (1979, 1981), Morgan (1983, 1986), Li (1983), and Li et al. (1987a, b, 1988) for more detailed description of the method.

Freudenstein and Roth (1963) introduced a numerical method called the *bootstrap method* or the *parameter perturbation method* for solving a very difficult problem associated with the coupler-point synthesis of geared five-bar linkages. This work attracted much attention from mathematicians and scientists and was subsequently refined and perfected into the modern continuation method (Saaty, 1981). Perhaps the two most important concepts in a successful implementation of the method are (1) the use of complex number to represent real parameters and variables, and (2) the introduction of m-homogeneous variables (Wampler et al., 1990). The homogeneous formulation may lead to a smaller number of solution paths and reduce the computing time needed in following some diverging paths.

In what follows, we first introduce the concept of polynomial homogenization and then the concept of homotopy. Tsai and Morgan's solution is used to demonstrate the methodology. Finally, a *cheater's homotopy* is introduced.

A.1 BEZOUT NUMBER

The Bezout number accounts for the number of solutions to a system of polynomial equations. First, we consider a single polynomial equation in one complex variable:

$$f(x) = \sum_{j=0}^{n} a_j x^{n-j} = 0, \tag{A.1}$$

where a_j are constant coefficients and x is a complex variable. A fundamental theorem of algebra states that by counting multiplicity of a solution, an nth-degree polynomial equation has n solutions. That is, the *Bezout number* of a single polynomial is equal to the degree of the equation.

Next, we consider a system of n polynomial equations in n variables. The degree of a multivariable polynomial term is defined as the sum of its exponents, the degree of a polynomial equation is defined as the highest degree of all the terms in the equation, and the *total degree* of a polynomial system is defined as the product of the degrees of all the equations. For the system of polynomials

$$\begin{aligned} f_1(x_1, x_2, \ldots, x_n) &= 0, \\ f_2(x_1, x_2, \ldots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, x_2, \ldots, x_n) &= 0, \end{aligned} \tag{A.2}$$

if the degree of equation $f_i(\mathbf{x})$ is d_i, the total degree of the system is $d = \prod_{i=1}^{n} d_i$.

The *Bezout theorem* states that a polynomial system of total degree d has at most d isolated solutions in the complex Euclidean space. If all solutions of a polynomial system are nonsingular, the number of solutions is exactly equal to the total degree of the system. Including solutions at infinity, the *Bezout number* of a polynomial system is equal to the total degree of the system.

A.2 TRADITIONAL HOMOGENEOUS FORMULATION

Solutions at infinity can properly be accounted for by introducing homogeneous coordinates. Let $(x_1, x_2, \ldots, x_n)$ be n variables of a polynomial system. The homogeneous coordinates $(y_1, y_2, \ldots, y_{n+1})$ are obtained by extending the number of variables from n to $n+1$ using the relations $x_i = y_i/y_{n+1}$, for $i = 1, 2, \ldots, n$, where y_{n+1} is a *scaling factor*. Substituting $x_i = y_i/y_{n+1}$

into the original system of equations and then multiplying each equation by $y_{n+1}^{d_j}$ to clear the denominators, we obtain a new system of equations that are homogeneous in y_i. In this way, solutions at infinity become finite nonzero solutions with $y_{n+1} = 0$, and a solution to the original system of equations is obtained by dividing the homogeneous solution by y_{n+1}. This is known as the traditional *1-homogeneous formulation.*

For example, the system of equations, Eqs. (2.106) through (2.110), derived for the inverse kinematics of a general $6R$ manipulator can be written as

$$F(\mathbf{x}) : \begin{cases} f_i(\mathbf{x}) = a_{i,1}x_1x_3 + a_{i,2}x_1x_4 + a_{i,3}x_2x_3 + a_{i,4}x_2x_4 \\ \qquad + a_{i,5}x_5x_7 + a_{i,6}x_5x_8 + a_{i,7}x_6x_7 + a_{i,8}x_6x_8 \\ \qquad + a_{i,9}x_1 + a_{i,10}x_2 + a_{i,11}x_3 + a_{i,12}x_4 \\ \qquad + a_{i,13}x_5 + a_{i,14}x_6 + a_{i,15}x_7 + a_{i,16}x_8 \\ \qquad + a_{i,17} = 0 \quad (i = 1, \dots, 4), \\ f_5(\mathbf{x}) = x_1^2 + x_2^2 - 1 = 0, \\ f_6(\mathbf{x}) = x_3^2 + x_4^2 - 1 = 0, \\ f_7(\mathbf{x}) = x_5^2 + x_6^2 - 1 = 0, \\ f_8(\mathbf{x}) = x_7^2 + x_8^2 - 1 = 0, \end{cases} \tag{A.3}$$

where $a_{i,j}$'s are the coefficients and x_i's are the variables of the eight polynomials. The coefficients represent the constant link parameters, and the variables are the sines and cosines of four joint angles. For this system of equations, the degree of $f_i(\mathbf{x}) = 0$ is 2 for $i = 1$ to 8. Hence the total degree of the system is $\prod_{i=1}^{8} d_i = 2^8 = 256$.

The traditional 1-homogeneous system is obtained by substituting $x_i = y_i/y_9$, for $i = 1, 2, \dots, 8$ into Eq. (A.3) and then multiplying each equation by y_9^2 to clear the denominators. This results in a system of eight homogeneous equations in nine unknowns:

$$\tilde{F}(\mathbf{y}) : \begin{cases} \tilde{f}_i(\mathbf{y}) = a_{i,1}y_1y_3 + a_{i,2}y_1y_4 + a_{i,3}y_2y_3 + a_{i,4}y_2y_4 \\ \qquad + a_{i,5}y_5y_7 + a_{i,6}y_5y_8 + a_{i,7}y_6y_7 + a_{i,8}y_6y_8 \\ \qquad + a_{i,9}y_1y_9 + a_{i,10}y_2y_9 + a_{i,11}y_3y_9 + a_{i,12}y_4y_9 \\ \qquad + a_{i,13}y_5y_9 + a_{i,14}y_6y_9 + a_{i,15}y_7y_9 + a_{i,16}y_8y_9 \\ \qquad + a_{i,17}y_9^2 = 0 \quad (i = 1, \dots, 4), \\ \tilde{f}_5(\mathbf{y}) = y_1^2 + y_2^2 - y_9^2 = 0, \\ \tilde{f}_6(\mathbf{y}) = y_3^2 + y_4^2 - y_9^2 = 0, \\ \tilde{f}_7(\mathbf{y}) = y_5^2 + y_6^2 - y_9^2 = 0, \\ \tilde{f}_8(\mathbf{y}) = y_7^2 + y_8^2 - y_9^2 = 0. \end{cases} \tag{A.4}$$

For a traditional 1-homogeneous system, if $(y_1, \ldots, y_{n+1})$ is a solution, $(\rho y_1, \ldots, \rho y_{n+1})$ is also a solution, where ρ is any nonzero constant. Hence each solution of the original system corresponds to a line passing the origin in the homogeneous coordinates called the *projective space*. Using homogeneous coordinates, the Bezout number of a system of polynomial equations is equal to the total degree of the system. Hence the Bezout number for the system of equations in Eq. (A.4) is 256.

A.3 *M*-HOMOGENIZATION

In this section we introduce the concept of *m-homogenization*. It has been shown that the use of multihomogeneous variables can sometimes reduce the number of extraneous solutions. The main tasks involved in multihomogenization are the association of variables in groups and the transformation of each group into a separate homogeneous set of variables. A polynomial system is said to be under *m-homogeneous formulation* if its variables are arranged into m homogeneous groups. For example, we may arrange the variables in the system of equations given by Eq. (A.3) into two groups: (x_1, x_2, x_5, x_6) and (x_3, x_4, x_7, x_8). Then we substitute $x_i = y_i/y_9$ for $i = 1$, 2, 5, and 6, and $x_i = y_i/y_{10}$ for $i = 3, 4, 7$, and 8 into the equations and clear the denominators to obtain a 2-homogeneous system:

$$\tilde{F}(\mathbf{y}) : \begin{cases} \tilde{f}_i(\mathbf{y}) = a_{i,1}y_1y_3 + a_{i,2}y_1y_4 + a_{i,3}y_2y_3 + a_{i,4}y_2y_4 \\ \qquad + a_{i,5}y_5y_7 + a_{i,6}y_5y_8 + a_{i,7}y_6y_7 + a_{i,8}y_6y_8 \\ \qquad + a_{i,9}y_1y_{10} + a_{i,10}y_2y_{10} + a_{i,11}y_3y_9 + a_{i,12}y_4y_9 \\ \qquad + a_{i,13}y_5y_{10} + a_{i,14}y_6y_{10} + a_{i,15}y_7y_9 + a_{i,16}y_8y_9 \\ \qquad + a_{i,17}y_9y_{10} = 0 \quad (i = 1, \ldots, 4), \\ \tilde{f}_5(\mathbf{y}) = y_1^2 + y_2^2 - y_9^2 = 0, \\ \tilde{f}_6(\mathbf{y}) = y_3^2 + y_4^2 - y_{10}^2 = 0, \\ \tilde{f}_7(\mathbf{y}) = y_5^2 + y_6^2 - y_9^2 = 0, \\ \tilde{f}_8(\mathbf{y}) = y_7^2 + y_8^2 - y_{10}^2 = 0. \end{cases} \tag{A.5}$$

Let β_j for $j = 1$ to m denote m groups of variables. To compute the Bezout number, we first form a sum over j of the product $d_{ij}\beta_j$ for each m-homogeneous equation, where d_{ij} is the degree of equation i with respect to the variables of the jth group. This results in n linear equations in β_j with d_{ij} as their coefficients. Next we form the product of these linear equations to

TABLE A.1. Degrees of the 2-Homogeneous System

Equation i	Group 1 ($j = 1$)	Group 2 ($j = 2$)
1	1	1
2	1	1
3	1	1
4	1	1
5	2	0
6	0	2
7	2	0
8	0	2

obtain a nonlinear polynomial in β_j:

$$\prod_{i=1}^{n}\left(\sum_{j=1}^{m} d_{ij}\beta_j\right) = (d_{11}\beta_1 + \cdots + d_{1m}\beta_m)(d_{21}\beta_1 + \cdots + d_{2m}\beta_m)$$
$$\cdots(d_{n1}\beta_1 + \cdots + d_{nm}\beta_m). \tag{A.6}$$

The *multihomogeneous Bezout number* is defined as the coefficient of the term $\prod_{j=1}^{m}\beta_j^{k_j}$ in Eq. (A.6), where k_j is the number of variables in group j.

For the 2-homogeneous system given in Eq. (A.5), the variables are arranged in two groups: (x_1, x_2, x_5, x_6) and (x_3, x_4, x_7, x_8). The degrees of the eight equations in the variables of each group are listed in Table A.1. To find the Bezout number, we form the product

$$\prod_{i=1}^{8}\left(\sum_{j=1}^{2} d_{ij}\beta_j\right) = (\beta_1 + \beta_2)^4(2\beta_1)^2(2\beta_2)^2,$$

where d_{ij} is taken from the (i, j) element of Table A.1. Since both groups contain four variables and the coefficient of the term $\beta_1^4\beta_2^4$ in the polynomial above is 96, the 2-homogeneous Bezout number is 96. Hence the 2-homogeneous formulation of the system results in at most 96 solutions.

A.4 SOLUTIONS AT INFINITY

As mentioned earlier, using the homogeneous coordinates, solutions at infinity can be accounted for by setting the last coordinate in each group of variables to zero. For example, solutions at infinity for the 2-homogeneous system given in Eq. (A.5) consist of three categories: (1) $y_9 = y_{10} = 0$, (2) $y_9 = 0$, $y_{10} = 1$, and (3) $y_9 = 1$, $y_{10} = 0$. For case (1), $y_9 = y_{10} = 0$, the last four equations of (A.5) reduce to

$$\begin{aligned}
\tilde{f}_5(\mathbf{y}) &= y_1^2 + y_2^2 = 0, \\
\tilde{f}_6(\mathbf{y}) &= y_3^2 + y_4^2 = 0, \\
\tilde{f}_7(\mathbf{y}) &= y_5^2 + y_6^2 = 0, \\
\tilde{f}_8(\mathbf{y}) &= y_7^2 + y_8^2 = 0.
\end{aligned} \tag{A.7}$$

Hence $y_2 = \pm i y_1$, $y_4 = \pm i y_3$, $y_6 = \pm i y_5$, and $y_8 = \pm i y_7$. There are 16 combinations. For a typical combination, say $y_2 = i y_1$, $y_4 = i y_3$, $y_6 = i y_5$, $y_8 = i y_7$, the first four equations of (A.5) reduce to

$$\begin{aligned}
\tilde{f}_i(\mathbf{y}) &= [(a_{i,1} - a_{i,4}) + i(a_{i,2} + a_{i,3})] y_1 y_3 \\
&\quad + [(a_{i,5} - a_{i,8}) + i(a_{i,6} + a_{i,7})] y_5 y_7 \\
&= 0,
\end{aligned} \tag{A.8}$$

for $i = 1, 2, 3$, and 4.

We may consider Eq. (A.8) as a system of four linear equations in two variables, $y_1 y_3$ and $y_5 y_7$. Then it becomes obvious that the only feasible solutions are $y_1 y_3 = 0$ and $y_5 y_7 = 0$ for generic parameters. However, y_1 and y_5 cannot be zero simultaneously, otherwise $(y_1, y_2, y_5, y_6, y_9) = (0, 0, 0, 0, 0)$, which is not in the projective space. Similarly, y_3 and y_7 cannot be zero simultaneously. Hence for this combination there are two solutions: $(y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8) = (1, i, 0, 0, 0, 0, 1, i)$ and $(0, 0, 1, i, 1, i, 0, 0)$. Since there are 16 combinations for case (1), a total of 32 isolated solutions exist at infinity.

Similarly, it can be shown that there are 16 solutions at infinity for each of cases (2) and (3). Overall, there are 64 solutions at infinity. Hence the system of equations has at most 32 solutions. Although these 32 solutions do satisfy Eq. (A.3), it can be shown that 16 of them do not satisfy the original loop-closure equations. Those solutions that do not satisfy the original loop-closure equations are called *extraneous solutions*, while the others are *significant*.

A.5 CONTINUATION METHOD

In this section we describe briefly a numerical solution method called the *continuation method*. Suppose that we would like to solve a system of n polynomial equations in n variables:

$$F(\mathbf{x}) : \begin{cases} f_1(x_1, x_2, \ldots, x_n) = 0, \\ f_2(x_1, x_2, \ldots, x_n) = 0, \\ \quad\vdots \\ f_n(x_1, x_2, \ldots, x_n) = 0. \end{cases} \tag{A.9}$$

We call $F(\mathbf{x}) = (f_1, f_2, \ldots, f_n)$ the *target system*. The idea of the continuation method is to embed the target system in a family of polynomial systems parameterized by a real parameter t, called the *continuation parameter*. All solutions to one system in the family, called the *initial system* $G(\mathbf{x}) = 0$, can be solved in closed form. Furthermore, the solutions of the initial system should be connected to all isolated solutions of the target system via smooth paths called the *homotopy paths*. More specifically, we construct a *homotopy function* $H(\mathbf{x}, t) = 0$ with a new parameter t such that $H(\mathbf{x}, 1) = F(\mathbf{x})$, $H(\mathbf{x}, 0) = G(\mathbf{x})$, and the following properties are satisfied:

1. *Triviality*. The solutions of $G(\mathbf{x}) = 0$ can be solved in closed form.
2. *Smoothness*. The solution set of $H(\mathbf{x}, t) = 0$ for $0 \leq t < 1$ consists of a finite number of smooth paths, each parameterized by t.
3. *Accessibility*. Every isolated solution of $F(\mathbf{x}) = 0$ is reached by a solution of $G(\mathbf{x}) = 0$.

Properties 1 to 3 ensure that the solution paths can be followed from the solutions of $G(\mathbf{x}) = 0$ at $t = 0$ to all solutions of $F(\mathbf{x}) = 0$ at $t = 1$ using a conventional numerical technique involving differential equation solvers. We note that even though properties 1 to 3 imply that each solution of $F(\mathbf{x}) = 0$ will lie at the end of a solution path, some of the paths may diverge to infinity as the continuation parameter t approaches 1. This usually occurs when $G(\mathbf{x}) = 0$ possesses more solutions than $F(\mathbf{x}) = 0$. Under such a condition, some of the solutions of $G(\mathbf{x}) = 0$ are extraneous and typically diverge to infinity.

We see that the continuation method requires an initial system $G(\mathbf{x}) = 0$, a homotopy function $H(\mathbf{x}, t) = 0$, and a numerical technique for tracking the homotopy function from $t = 0$ to $t = 1$. In what follows, we briefly describe each of these in turn.

A.5.1 Homotopy Function

The homotopy function provides a way of tracking the solutions of an initial system to the solutions of a target system. The solution paths must be smooth and nonsingular. A commonly used generic homotopy function is

$$H(\mathbf{x}, t) = \gamma(1 - t)G(\mathbf{x}) + tF(\mathbf{x}), \tag{A.10}$$

where γ is a random complex constant. We note that at $t = 0$ the solutions of $H(\mathbf{x}, 0) = 0$ are those of $G(\mathbf{x}) = 0$ and at $t = 1$ its solutions are those of $F(\mathbf{x}) = 0$.

In constructing a homotopy function, the triviality property can easily be satisfied. However, the smoothness and accessibility properties can only be satisfied with a careful choice of $G(\mathbf{x})$ and γ. The parameter γ is needed to ensure the accessibility of a homotopy function. This can be demonstrated by an example in which the initial system is the negative of the original system; that is, $G(\mathbf{x}) = -F(\mathbf{x})$. For $\gamma = 1$, the homotopy function $H(\mathbf{x}, t) = (1 - t)G(\mathbf{x}) + tF(\mathbf{x})$ would vanish at $t = 0.5$, while for other choices of γ, it would not. Therefore, it is necessary to choose γ at random to avoid this problem.

A.5.2 Initial System

All the solutions of an initial system must be known, each solution must be nonsingular, and the system must maintain the same *polynomial structure* as the target system. Otherwise, several things can go wrong: (1) $H(\mathbf{x}, t) = 0$ might have no solution for some increment of t, or it might have several nearby solutions which are hard to distinguish or find; (2) $G(\mathbf{x}) = 0$ might not have enough solutions to match the solutions of $F(\mathbf{x}) = 0$, or $F(\mathbf{x}) = 0$ might have some solutions with no continuation paths converging on them; and (3) as t approaches 1, the solutions of $H(\mathbf{x}, t) = 0$ might diverge to infinity.

Perhaps the simplest initial system for a 1-homogeneous polynomial system is (Li, 1983)

$$G(\mathbf{x}) : \begin{cases} g_1(\mathbf{x}) = a_1 x_1^{d_1} - b_1 = 0, \\ g_2(\mathbf{x}) = a_2 x_2^{d_2} - b_2 = 0, \\ \quad\vdots \\ g_n(\mathbf{x}) = a_n x_n^{d_n} - b_n = 0, \end{cases} \tag{A.11}$$

where a_i and b_i are random complex coefficients. Because of the simple structure of $G(\mathbf{x})$ in Eq. (A.11), the triviality property is clearly satisfied. It can be shown that the smoothness and accessibility properties are satisfied for almost all choices of a_i and b_i. That is, $G(\mathbf{x}) = 0$ consists of $d = \prod_{i=1}^{n} d_i$ smooth paths, which either converge to the solutions of $F(\mathbf{x}) = 0$ or diverge to infinity as t approaches 1, and each isolated solution of $F(\mathbf{x}) = 0$ has a path converging to it. Hence the only bad behavior is that as t is incremented, the solution to $H(\mathbf{x}, t) = 0$ might diverge to infinity. Nevertheless, all isolated solutions of $F(\mathbf{x}) = 0$ can be found. The random choice of a_j and b_j is a key factor for the success of this method.

Another possible generic initial system is

$$G(\mathbf{x}) : \begin{cases} g_1(\mathbf{x}) = \prod_{j=1}^{d_1}(x_1 - b_{1j}) = 0, \\ g_2(\mathbf{x}) = \prod_{j=1}^{d_2}(x_2 - b_{2j}) = 0, \\ \quad\vdots \\ g_n(\mathbf{x}) = \prod_{j=1}^{d_n}(x_n - b_{nj}) = 0, \end{cases} \tag{A.12}$$

where b_{ij} are random complex coefficients. Again it can be shown that Eq. (A.12) satisfies the triviality, smoothness, and accessibility properties for almost all choices b_{ij}.

The foregoing initial systems are generic in nature and can be applied to any polynomial system. It is also possible to construct an initial system for a specific problem of interest. A straightforward approach is to randomly perturb the coefficients of a target system without damaging the basic structure of the polynomials. For an m-homogeneous system, an initial system with identical polynomial structure as the target system can often be generated by a product of factors:

$$g_i(\mathbf{x}) = \prod_{j=1}^{m} h_{ij}(x_{1j}, x_{2j}, \ldots, x_{k_j j}) = 0, \tag{A.13}$$

where h_{ij} is a polynomial of degree d_{ij} with respect to group j, d_{ij} is the degree of equation i with respect to the variables of group j, and k_j denotes the number of variables in group j. The factors h_{ij} should be chosen with sufficiently random coefficients to satisfy the triviality, smoothness, and accessibility properties.

For example, an initial system for the 2-homogeneous system given in Eq. (A.5) can be formulated as

$$G(\mathbf{y}):\begin{cases} g_1(\mathbf{y}) = (y_1 - b_{11}y_9)(y_3 - b_{12}y_{10}) = 0, \\ g_2(\mathbf{y}) = (y_2 - b_{21}y_9)(y_4 - b_{22}y_{10}) = 0, \\ g_3(\mathbf{y}) = (y_5 - b_{31}y_9)(y_7 - b_{32}y_{10}) = 0, \\ g_4(\mathbf{y}) = (y_6 - b_{41}y_9)(y_8 - b_{42}y_{10}) = 0, \\ g_5(\mathbf{y}) = y_1^2 + y_2^2 - b_{51}y_9^2 = 0, \\ g_6(\mathbf{y}) = y_3^2 + y_4^2 - b_{61}y_{10}^2 = 0, \\ g_7(\mathbf{y}) = y_5^2 + y_6^2 - b_{71}y_9^2 = 0, \\ g_8(\mathbf{y}) = y_7^2 + y_8^2 - b_{81}y_{10}^2 = 0, \end{cases} \tag{A.14}$$

where the b_{ij} are random complex constants.

A.5.3 Path Tracking

To implement the continuation method, we start at a solution of $G(\mathbf{x}) = 0$, increment t by a small number δt and solve for the function $H(\mathbf{x}, \delta t) = 0$. Increment t again and solve for $H(\mathbf{x}, 2\delta t) = 0$, and so on, until t reaches $t = 1$. Each time we increment t by δt, we use the solution of $H(\mathbf{x}, t) = 0$ obtained at the preceding step as the initial guess and solve for the function $H(\mathbf{x}, t + \delta t) = 0$. We repeat this process for each solution of $G(\mathbf{x}) = 0$ to find all solutions of $F(\mathbf{x}) = 0$.

The method can be implemented on a computer. The implementation involves the formulation of a set of differential equations whose solutions are the continuation paths. Typically, these differential equations are solved by using a numerical integration technique. The integration is followed by Newton's iteration to refine the final solution estimates at the end of each solution path (Morgan, 1983, 1986).

A.5.4 Tsai and Morgan's Solution

As an example, let us consider the inverse kinematics of the general $6R$ manipulator studied in Chapter 2. Expanding Eqs. (2.106) through (2.110), we obtain a system of eight polynomials as shown in Eq. (A.15), where $x_1 = c\theta_1, x_2 = s\theta_1, x_3 = c\theta_2, x_4 = s\theta_2, x_5 = c\theta_4, x_6 = s\theta_4, x_7 = c\theta_5$, and $x_8 = s\theta_5$ are the variables of the eight polynomials; a_i, d_i, and α_i are the constant D-H link parameters; $\theta_i, i = 1, 2, 4$, and 5 are the joint angles; $\mathbf{p} = [p_x, p_y, p_z]^{\mathrm{T}}$ is a position vector of the origin of the fifth link frame; and $\mathbf{e} = [e_x, e_y, e_z]^{\mathrm{T}}$ is a unit vector along the z_5-axis, as shown in Fig. 2.11. The position vector $\mathbf{p}$ and the unit vector $\mathbf{e}$ are computed by Eqs. (2.100) and (2.101) using the given end-effector position and orientation.

$$
F(\mathbf{x}):\begin{cases}
f_1(\mathbf{x}) = & -c\alpha_1 s\alpha_2 p_y x_1 x_3 + s\alpha_2 p_x x_1 x_4 + c\alpha_1 s\alpha_2 p_x x_2 x_3 + s\alpha_2 p_y x_2 x_4 \\
& - s\alpha_3 c\alpha_4 a_5 x_5 x_8 - s\alpha_3 a_5 x_6 x_7 - s\alpha_1 c\alpha_2 p_y x_1 + s\alpha_1 c\alpha_2 p_x x_2 \\
& - s\alpha_1 s\alpha_2 (p_z - d_1) x_3 - s\alpha_2 a_1 x_4 + s\alpha_3 s\alpha_4 d_5 x_5 - s\alpha_3 a_4 x_6 \\
& - c\alpha_3 s\alpha_4 a_5 x_8 + (c\alpha_1 c\alpha_2 p_z - c\alpha_1 c\alpha_2 d_1 - c\alpha_2 d_2 - d_3 \\
& - c\alpha_3 d_4 - c\alpha_3 c\alpha_4 d_5) = 0, \\
f_2(\mathbf{x}) = & -c\alpha_1 s\alpha_2 e_y x_1 x_3 + s\alpha_2 e_x x_1 x_4 + c\alpha_1 s\alpha_2 e_x x_2 x_3 + s\alpha_2 e_y x_2 x_4 \\
& + s\alpha_3 c\alpha_4 s\alpha_5 x_5 x_7 - s\alpha_3 s\alpha_5 x_6 x_8 - s\alpha_1 c\alpha_2 e_y x_1 + s\alpha_1 c\alpha_2 e_x x_2 \\
& - s\alpha_1 s\alpha_2 e_z x_3 + s\alpha_3 s\alpha_4 c\alpha_5 x_5 + c\alpha_3 s\alpha_4 s\alpha_5 x_7 \\
& + (c\alpha_1 c\alpha_2 e_z - c\alpha_3 c\alpha_4 c\alpha_5) = 0, \\
f_3(\mathbf{x}) = & a_2 e_x x_1 x_3 + c\alpha_1 a_2 e_y x_1 x_4 + a_2 e_y x_2 x_3 - c\alpha_1 a_2 e_x x_2 x_4 \\
& - s\alpha_3 c\alpha_4 s\alpha_5 d_3 x_5 x_7 + s\alpha_5 a_3 x_5 x_8 + c\alpha_4 s\alpha_5 a_3 x_6 x_7 \\
& + s\alpha_3 s\alpha_5 d_3 x_6 x_8 + (a_1 e_x - s\alpha_1 d_2 e_y) x_1 + (a_1 e_y + s\alpha_1 d_2 e_x) x_2 \\
& + s\alpha_1 a_2 e_z x_4 - s\alpha_3 s\alpha_4 c\alpha_5 d_3 x_5 + s\alpha_4 c\alpha_5 a_3 x_6 \\
& - s\alpha_4 s\alpha_5 (d_4 + c\alpha_3 d_3) x_7 + s\alpha_5 a_4 x_8 + (d_1 e_z + c\alpha_1 d_2 e_z \\
& + c\alpha_3 c\alpha_4 c\alpha_5 d_3 + c\alpha_4 c\alpha_5 d_4 + c\alpha_5 d_5 - p_x e_x - p_y e_y \\
& - p_z e_z) = 0, \\
f_4(\mathbf{x}) = & a_2 p_x x_1 x_3 + c\alpha_1 a_2 p_y x_1 x_4 + a_2 p_y x_2 x_3 - c\alpha_1 a_2 p_x x_2 x_4 \\
& + a_3 a_5 x_5 x_7 + s\alpha_3 c\alpha_4 a_5 d_3 x_5 x_8 + s\alpha_3 a_5 d_3 x_6 x_7 - c\alpha_4 a_3 a_5 x_6 x_8 \\
& + (a_1 p_x - s\alpha_1 d_2 p_y) x_1 + (a_1 p_y + s\alpha_1 d_2 p_x) x_2 - a_1 a_2 x_3 \\
& - s\alpha_1 a_2 (d_1 - p_z) x_4 + (a_3 a_4 - s\alpha_3 s\alpha_4 d_3 d_5) x_5 \\
& + (s\alpha_4 a_3 d_5 + s\alpha_3 a_4 d_3) x_6 + a_4 a_5 x_7 + (s\alpha_4 a_5 d_4 \\
& + c\alpha_3 s\alpha_4 a_5 d_3) x_8 + 0.5(-a_1^2 - d_1^2 - a_2^2 - d_2^2 + a_3^2 + d_3^2 \\
& + a_4^2 + d_4^2 + a_5^2 + d_5^2 - p_x^2 - p_y^2 - p_z^2) = 0, \\
f_5(\mathbf{x}) = & x_1^2 + x_2^2 - 1 = 0, \\
f_6(\mathbf{x}) = & x_3^2 + x_4^2 - 1 = 0, \\
f_7(\mathbf{x}) = & x_5^2 + x_6^2 - 1 = 0, \\
f_8(\mathbf{x}) = & x_7^2 + x_8^2 - 1 = 0,
\end{cases} \tag{A.15}
$$

Tsai and Morgan (1985) used Eq. (A.10) with $\gamma = 1$ as the homotopy function to solve for the system of equations above. The D-H parameters of a $6R$ manipulator are given in Table A.2, and the given end-effector position and orientation are listed in Table A.3. Using Eq. (A.11) as the initial system, there are 256 generic starting points. A continuation path is generated for each

TABLE A.2. D-H Parameters of a $6R$ Manipulator

i	a_i (unit)	d_i (unit)	α_i (deg)
1	0.5000	0.1875	80
2	1.0000	0.375	15
3	0.1250	0.250	120
4	0.6250	0.875	75
5	0.3125	0.500	100
6	0.2500	0.125	60

TABLE A.3. Given End-Effector Position and Orientation

Given Vector	x-Component	y-Component	z-Component
q	0.22441776	0.71549788	0.79551628
u	−0.71511545	0.65150320	0.25328538
v	−0.69899036	−0.66895464	−0.25280857
w	0.00473084	−0.35783135	0.93377425

starting point. Some of these paths converge to solutions of Eq. (A.15), and the remaining diverge to infinity as the continuation parameter t approaches $t = 1$. Each isolated solution to the system has a path converging to it. All solutions are isolated unless the system has an infinite number of solutions, that is, when the manipulator is in a singular configuration. In the latter case, some solutions will be isolated and others will form algebraic hypersurfaces, most commonly curves. Hence the continuation method will find all solutions unless there are an infinite number of solutions. For the general $6R$ manipulator, the continuation method always has 32 converging paths, of which 16 are significant. For the example above, the continuation method found 12 significant real solutions listed in Table A.4.

Following Eq. (A.13), an alternative initial system can be constructed:

$$G(\mathbf{x}) : \begin{cases} g_1(\mathbf{x}) = -x_6x_7 + x_2 = 0, \\ g_2(\mathbf{x}) = -x_6x_8 + x_2 = 0, \\ g_3(\mathbf{x}) = x_1x_3 + x_5x_8 + x_1 + x_2 - x_7 - 5 = 0, \\ g_4(\mathbf{x}) = x_1x_3 + x_5x_7 + x_1 + x_2 - x_3 + x_8 - 5 = 0, \\ g_5(\mathbf{x}) = x_1^2 + x_2^2 - 1 = 0, \\ g_6(\mathbf{x}) = x_3^2 + x_4^2 - 1 = 0, \\ g_7(\mathbf{x}) = x_5^2 + x_6^2 - 1 = 0, \\ g_8(\mathbf{x}) = x_7^2 + x_8^2 - 2 = 0. \end{cases} \tag{A.16}$$

It can be shown that the system above has exactly 32 isolated solutions, which can easily be obtained by straightforward elimination. Using a generic homotopy function, every isolated solution of the system of equations (A.15) can be reached by a solution of Eq. (A.16). Since there are exactly 32 paths to follow, this results in a great saving of computation time. The algorithm was implemented by Li and Wang (1990) and executed without any failure for various sets of parameters.

TABLE A.4. Real Solutions Found

No.	θ_1 (deg.)	θ_2 (deg.)	θ_4 (deg.)	θ_5 (deg)
1	167.68	83.55	65.84	−88.67
2	−143.00	100.07	18.46	−59.49
3	115.86	−168.65	157.17	−111.41
4	107.56	2.00	166.77	−173.54
5	−106.07	−140.86	−161.28	35.54
6	−65.37	142.24	−70.90	−51.63
7	120.52	31.27	114.15	−143.62
8	7.75	103.87	−21.37	−79.90
9	−16.69	97.90	−80.98	−25.72
10	47.26	163.44	28.32	−41.13
11	20.93	58.74	−27.07	−125.66
12	38.93	−56.45	12.28	72.23

A.6 CHEATER'S HOMOTOPY

The *cheater's homotopy* was introduced for solving a system of polynomial equations which needs to be solved repetitively with varying coefficients (Li et al., 1988). Let

$$F(\mathbf{x}, \mathbf{a}) : \begin{cases} f_1(x_1, \ldots, x_n; a_1, \ldots, a_k) = 0, \\ f_2(x_1, \ldots, x_n; a_1, \ldots, a_k) = 0, \\ \vdots \\ f_n(x_1, \ldots, x_n; a_1, \ldots, a_k) = 0 \end{cases} \tag{A.17}$$

be the target system, where $\mathbf{x} = (x_1, \ldots, x_n)$ are the variables and $\mathbf{a} = (a_1, \ldots, a_k)$ are the coefficients. Also let

$$G(\mathbf{x}) : \begin{cases} g_1(x_1, \ldots, x_n) = f_1(x_1, \ldots, x_n; a_1^0, \ldots, a_k^0) + b_1^0 = 0, \\ g_2(x_1, \ldots, x_n) = f_2(x_1, \ldots, x_n; a_1^0, \ldots, a_k^0) + b_2^0 = 0, \\ \vdots \\ g_n(x_1, \ldots, x_n) = f_n(x_1, \ldots, x_n; a_1^0, \ldots, a_k^0) + b_n^0 = 0 \end{cases} \tag{A.18}$$

be a system of polynomial equations obtained by substituting the coefficients $\mathbf{a}$ of the original system with the random complex numbers $\mathbf{a}^0 = (a_1^0, a_2^0, \ldots, a_k^0)$ and by adding the constants $\mathbf{b}^0 = (b_1^0, b_2^0, \ldots, b_n^0)$. It can be shown that the smoothness and accessibility properties hold for the homotopy function

$$H(\mathbf{x}, t) = F\big(\mathbf{x}, (1-t)\mathbf{a}^0 + t\mathbf{a}\big) + (1-t)\mathbf{b}^0 = 0. \tag{A.19}$$

Note that at $t = 0$, $H(\mathbf{x}, 0) = G(\mathbf{x}) = 0$, and at $t = 1$, $H(\mathbf{x}, 1) = F(\mathbf{x}, \mathbf{a}) = 0$.

The cheater's homotopy is accomplished in two steps.

1. Solve the system of equations in Eq. (A.18) using conventional homotopy. The number of nonsingular solutions found, d_0, is bounded by the total degree of the original system (i.e., $d_0 \leq d$).
2. For each new choice of coefficients $\mathbf{a} = (a_1, a_2, \ldots, a_k)$, follow the d_0 paths defined by $H(\mathbf{x}, t) = 0$ in Eq. (A.19) to find all solutions of $F(\mathbf{x}, \mathbf{a}) = 0$.

Using this method, we follow all $d = d_1 d_2 \cdots d_n$ paths of the system of equations with random complex coefficients $\mathbf{a}^0$ and $\mathbf{b}^0$ only once to find d_0 nonsingular solutions. These d_0 solutions are then used to initialize the paths for subsequent runs. This results in a fewer number of paths to follow and therefore drastically increases the efficiency of the method.

It is important for $\mathbf{a}^0$ and $\mathbf{b}^0$ to be complex numbers, although the original system $F(\mathbf{x}, \mathbf{a}) = 0$ may have real coefficients. For real $\mathbf{a}^0$ and $\mathbf{b}^0$, the homotopy function may fail to satisfy the smoothness and accessibility properties. Freudenstein and Roth (1963) did not use complex numbers to represent real parameters in their original *bootstrap method.* As a result, they encountered some difficulties in finding the solutions. The cheater's homotopy is called the *parameter homotopy* by Wampler et al. (1990).

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APPENDIX B

SYLVESTER DIALYTIC ELIMINATION METHOD

In this appendix we introduce an elimination method for reduction of a system of polynomial equations. The elimination method was introduced by Cayley (1848). The method requires the derivation of an equation known as the *eliminant* or *resultant*. Various methods of formulating the eliminant can be found in Salmon (1964). Perhaps the most widely used method in the field of kinematics is the *Sylvester dialytic elimination method* (Roth, 1993). Theoretically, the method will reduce any system of multivariable polynomial equations to a single polynomial in one unknown. Practically, the method can be applied only to a relatively small set of polynomial equations. This is because the resulting polynomial equation can explode exponentially with the number and degree of equations and can possibly introduce a large number of extraneous solutions.

B.1 ELIMINATION PROCEDURE

The Sylvester dialytic elimination procedure consists of the following six basic steps:

1. Rewrite all equations with one variable suppressed.
2. Define the power products of the remaining variables as new "linear unknowns." Here *power products* refers to the terms in a polynomial equation. For example, the power products of the polynomial $x^2z + 6xy + 3yz + 4z + 1 = 0$ are x^2z, xy, yz, z, and 1.

3. From the original equations, generate as many new linearly independent equations as the number of the linear unknowns.
4. Set the determinant of the coefficient matrix to zero to obtain a polynomial in the suppressed variable.
5. Find the roots of the characteristic polynomial of the matrix. This results in all possible solutions of the suppressed variable.
6. Substitute the suppressed variable, one at a time, into the original system of equations and repeat the process for the remaining variables.

Perhaps the most crucial step of the elimination procedure is step 3. The new linear equations are usually generated by multiplying the original equations by the powers of one or more of the variables (Salmon, 1964). This can best be illustrated by an example.

B.2 EXAMPLE

We wish to solve the following two polynomial equations in two unknowns, x_1 and x_2:

$$e_{11}x_1^2 + e_{12}x_2^2 + e_{13}x_1x_2 + e_{14}x_1 + e_{15}x_2 + e_{16} = 0, \tag{B.1}$$

$$e_{21}x_1^2 + e_{22}x_2^2 + e_{23}x_1x_2 + e_{24}x_1 + e_{25}x_2 + e_{26} = 0, \tag{B.2}$$

where e_{ij}'s are constant coefficients.

Step 1. We rewrite Eqs. (B.1) and (B.2) with the variable x_2 suppressed:

$$Ax_1^2 + Bx_1 + C = 0, \tag{B.3}$$

$$A'x_1^2 + B'x_1 + C' = 0, \tag{B.4}$$

where $A = e_{11}$, $B = e_{13}x_2 + e_{14}$, $C = e_{12}x_2^2 + e_{15}x_2 + e_{16}$, $A' = e_{21}$, $B' = e_{23}x_2 + e_{24}$, and $C' = e_{22}x_2^2 + e_{25}x_2 + e_{26}$.

Step 2. We consider the power products x_1^2, x_1, and 1 as new linear variables. Then Eqs. (B.3) and (B.4) become two linear equations in these new variables.

Step 3. We generate additional equations by multiplying Eqs. (B.3) and (B.4) by the powers of the variable x_1. Multiplying Eqs. (B.3) and (B.4) by x_1, we obtain

$$Ax_1^3 + Bx_1^2 + Cx_1 = 0, \tag{B.5}$$

$$A'x_1^3 + B'x_1^2 + C'x_1 = 0. \tag{B.6}$$

Step 4. We consider Eqs. (B.3), (B.4), (B.5), and (B.6) as four linear equations in four unknowns: x_1^3, x_1^2, x_1, and 1. Setting the determinant of the coefficient matrix of the four linearly independent equations to zero yields an eliminant:

$$\begin{vmatrix} A & B & C & 0 \\ 0 & A & B & C \\ A' & B' & C' & 0 \\ 0 & A' & B' & C' \end{vmatrix} = 0. \tag{B.7}$$

Expanding Eq. (B.7), we obtain

$$(AC' - CA')^2 + (AB' - BA')(CB' - BC') = 0. \tag{B.8}$$

Step 5. Equation (B.8) is a fourth-degree polynomial in x_2. Hence, counting multiplicities and solutions at infinity, there are exactly four solutions.

Step 6. We substitute each solution of x_2 into Eqs. (B.1) and (B.2) and solve the resulting equations for x_1. There should be only one solution of x_1 corresponding to each solution of x_2, since Eqs. (B.1) and (B.2) must be satisfied simultaneously.

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APPENDIX C

RAGHAVAN AND ROTH'S SOLUTION

The Sylvester dialytic elimination method described in Appendix B is effective for a small system of polynomial equations of relatively low degree. For more complicated polynomial systems, the method can make a problem unmanageably large. Therefore, it is desirable to develop alternative methods of generating new linearly independent equations that introduce no new power products, or at worst a small number of new power products. In this appendix we describe a technique employed by Raghavan and Roth (1990a,b) for solving the inverse kinematics of the general $6R$ manipulator.

C.1 LOOP-CLOSURE EQUATION

Let us consider the general $6R$ manipulator shown in Fig. 2.11. For convenience, we decompose ${}^1A_2$ as a product of two matrices:

$$ {}^1A_2 = {}^1G_2 \, {}^1H_2, \tag{C.1} $$

where

$$ {}^1G_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad {}^1H_2 = \begin{bmatrix} 1 & 0 & 0 & a_2 \\ 0 & c\alpha_2 & -s\alpha_2 & 0 \\ 0 & s\alpha_2 & c\alpha_2 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}. $$

Note that ${}^1G_2$ contains only the joint variable and ${}^1H_2$ contains only the link parameters.

Substituting Eq. (C.1) into (2.99), we obtain

$$ {}^0A_1\, {}^1G_2\, {}^1H_2\, {}^2A_3\, {}^3A_4\, {}^4A_5\, {}^5A_6 = {}^0A_6. \tag{C.2} $$

Premultipling both sides of Eq. (C.2) by $({}^0A_1\, {}^1G_2)^{-1}$ and postmultiplying both sides by ${}^5A_6^{-1}$, we obtain

$$ {}^1H_2\, {}^2A_3\, {}^3A_4\, {}^4A_5 = {}^1G_2^{-1}\, {}^0A_1^{-1}\, {}^0A_6\, {}^5A_6^{-1}. \tag{C.3} $$

Note that by moving θ_1, θ_2, and θ_6 to the right-hand side of the equation, we have effectively lowered the degrees of the equations. When the matrix multiplication is carried out, Eq. (C.3) takes the form

$$ \begin{bmatrix} f_{11}(\theta_3,\theta_4,\theta_5) & f_{12}(\theta_3,\theta_4,\theta_5) & f_{13}(\theta_3,\theta_4,\theta_5) & f_{14}(\theta_3,\theta_4,\theta_5) \\ f_{21}(\theta_3,\theta_4,\theta_5) & f_{22}(\theta_3,\theta_4,\theta_5) & f_{23}(\theta_3,\theta_4,\theta_5) & f_{24}(\theta_3,\theta_4,\theta_5) \\ f_{31}(\theta_4,\theta_5) & f_{32}(\theta_4,\theta_5) & f_{33}(\theta_4,\theta_5) & f_{34}(\theta_4,\theta_5) \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ = \begin{bmatrix} f'_{11}(\theta_1,\theta_2,\theta_6) & f'_{12}(\theta_1,\theta_2,\theta_6) & f'_{13}(\theta_1,\theta_2) & f'_{14}(\theta_1,\theta_2) \\ f'_{21}(\theta_1,\theta_2,\theta_6) & f'_{22}(\theta_1,\theta_2,\theta_6) & f'_{23}(\theta_1,\theta_2) & f'_{24}(\theta_1,\theta_2) \\ f'_{31}(\theta_1,\theta_2,\theta_6) & f'_{32}(\theta_1,\theta_2,\theta_6) & f'_{33}(\theta_1,\theta_2) & f'_{14}(\theta_1,\theta_2) \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{C.4} $$

Equation (C.4) only reveals the variables appearing in the elements of Eq. (C.3). An examination of Eq. (C.4) reveals that the six scalar equations obtained from the third and fourth columns are free of the variable θ_6. These six equations can be written in vector form, denoted as **a** and **b**, as follows:

$$ \mathbf{a}: \begin{bmatrix} c\theta_3 & s\theta_3 & 0 \\ -c\alpha_2 s\theta_3 & c\alpha_2 c\theta_3 & s\alpha_2 \\ s\alpha_2 s\theta_3 & -s\alpha_2 c\theta_3 & c\alpha_2 \end{bmatrix} \begin{bmatrix} \mu_x \\ \mu_y \\ \mu_z \end{bmatrix} + \begin{bmatrix} a_2 \\ 0 \\ d_2 \end{bmatrix} = \begin{bmatrix} c\theta_2 & s\theta_2 & 0 \\ s\theta_2 & -c\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix}, \tag{C.5} $$

$$ \mathbf{b}: \begin{bmatrix} c\theta_3 & s\theta_3 & 0 \\ -c\alpha_2 s\theta_3 & c\alpha_2 c\theta_3 & s\alpha_2 \\ s\alpha_2 s\theta_3 & -s\alpha_2 c\theta_3 & c\alpha_2 \end{bmatrix} \begin{bmatrix} \nu_x \\ \nu_y \\ \nu_z \end{bmatrix} = \begin{bmatrix} c\theta_2 & s\theta_2 & 0 \\ s\theta_2 & -c\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}, \tag{C.6} $$

where

$$ \mu_x = g_x c\theta_4 + g_y s\theta_4 + a_3, $$

$$ \mu_y = -(g_x s\theta_4 - g_y c\theta_4) c\alpha_3 + g_z s\alpha_3, $$

$$ \mu_z = (g_x s\theta_4 - g_y c\theta_4) s\alpha_3 + g_z c\alpha_3 + d_3, $$

$$\nu_x = m_x c\theta_4 + m_y s\theta_4,$$

$$\nu_y = -(m_x s\theta_4 - m_y c\theta_4)c\alpha_3 + m_z s\alpha_3,$$

$$\nu_z = (m_x s\theta_4 - m_y c\theta_4)s\alpha_3 + m_z c\alpha_3,$$

and where g_x, g_y, g_z, h_x, h_y, h_z, m_x, m_y, m_z, n_x, n_y, and n_z are defined in Chapter 2 under Eqs. (2.106) and (2.107). We note that μ_x, μ_y, μ_z, ν_x, ν_y, and ν_z are linear functions of the terms $s\theta_4 s\theta_5$, $s\theta_4 c\theta_5$, $c\theta_4 s\theta_5$, $c\theta_4 c\theta_5$, $s\theta_4$, $c\theta_4$, $s\theta_5$, $c\theta_5$, and 1, whereas h_x, h_y, h_z, n_x, n_y, and n_z are linear functions of the terms $s\theta_1$, $c\theta_1$, and 1.

The two vectors **a** and **b** in Eqs. (C.5) and (C.6) represent six scalar equations in five unknowns: $\theta_1, \theta_2, \ldots, \theta_5$. To eliminate several variables at a time, we treat some of the power products as new variables with the other power products suppressed. Toward this end, we write Eqs. (C.5) and (C.6) in the matrix form

$$A \begin{bmatrix} s\theta_4 s\theta_5 \\ s\theta_4 c\theta_5 \\ c\theta_4 s\theta_5 \\ c\theta_4 c\theta_5 \\ s\theta_4 \\ c\theta_4 \\ s\theta_5 \\ c\theta_5 \\ 1 \end{bmatrix} = B \begin{bmatrix} s\theta_1 s\theta_2 \\ s\theta_1 c\theta_2 \\ c\theta_1 s\theta_2 \\ c\theta_1 c\theta_2 \\ s\theta_1 \\ c\theta_1 \\ s\theta_2 \\ c\theta_2 \end{bmatrix}, \tag{C.7}$$

where A is a 6×9 matrix whose elements are linear combinations of $s\theta_3$, $c\theta_3$, and 1, and B is a 6×8 matrix whose elements are all constants.

It can be shown that the products $\mathbf{a}^T\mathbf{a}$, $\mathbf{a}^T\mathbf{b}$, $\mathbf{a} \times \mathbf{b}$, and $(\mathbf{a}^T\mathbf{a})\mathbf{b} - 2(\mathbf{a}^T\mathbf{b})\mathbf{a}$ result in eight additional polynomials which take the same form as Eq. (C.7) (Raghavan and Roth, 1990a). Combining these eight equations with Eq. (C.7), we obtain 14 linearly independent equations, which can be written

$$A' \begin{bmatrix} s\theta_4 s\theta_5 \\ s\theta_4 c\theta_5 \\ c\theta_4 s\theta_5 \\ c\theta_4 c\theta_5 \\ s\theta_4 \\ c\theta_4 \\ s\theta_5 \\ c\theta_5 \\ 1 \end{bmatrix} = B' \begin{bmatrix} s\theta_1 s\theta_2 \\ s\theta_1 c\theta_2 \\ c\theta_1 s\theta_2 \\ c\theta_1 c\theta_2 \\ s\theta_1 \\ c\theta_1 \\ s\theta_2 \\ c\theta_2 \end{bmatrix}, \tag{C.8}$$

where A' is a 14×9 matrix whose elements are linear combinations of $s\theta_3$, $c\theta_3$, and 1, and B' is a 14×8 matrix whose elements are constants.

C.2 ELIMINATION OF θ_1 AND θ_2

In this section we show how θ_1 and θ_2 can be eliminated simultaneously from Eq. (C.8). Toward this goal, we treat $s\theta_1 s\theta_2$, $s\theta_1 c\theta_2$, $c\theta_1 s\theta_2$, $c\theta_1 c\theta_2$, $s\theta_1$, $c\theta_1$, $s\theta_2$, and $c\theta_2$ in Eq. (C.8) as eight independent variables, and the left-hand-side terms as constants. Then Eq. (C.8) represents 14 linearly independent equations in eight unknowns. We can solve these eight variables from any eight of the 14 equations and substitute them back into the remaining six equations. This results in six independent equations, free of θ_1 and θ_2, which can be arranged in matrix form:

$$E \begin{bmatrix} s\theta_4 s\theta_5 \\ s\theta_4 c\theta_5 \\ c\theta_4 s\theta_5 \\ c\theta_4 c\theta_5 \\ s\theta_4 \\ c\theta_4 \\ s\theta_5 \\ c\theta_5 \\ 1 \end{bmatrix} = [0], \tag{C.9}$$

where E is a 6×9 matrix whose elements are linear combinations of $s\theta_3$, $c\theta_3$, and 1.

C.3 ELIMINATION OF θ_4 AND θ_5

In this section we eliminate θ_4 and θ_5 simultaneously. We note that the six equations in Eq. (C.9) are already written with the variable θ_3 suppressed. We make use of the following trigonometric identities to convert the equations into polynomials.

$$c\theta_i = \frac{1 - t_i^2}{1 + t_i^2}, \tag{C.10}$$

$$s\theta_i = \frac{2t_i}{1 + t_i^2}, \tag{C.11}$$

where $t_i = \tan(\theta_i/2)$.

Substituting Eqs. (C.10) and (C.11) for $i = 4$ and 5 into Eq. (C.9) and then multiplying each equation by $(1 + t_4^2)(1 + t_5^2)$ to clear the denominators, we obtain

$$E' \begin{bmatrix} t_4^2 t_5^2 \\ t_4^2 t_5 \\ t_4^2 \\ t_4 t_5^2 \\ t_4 t_5 \\ t_4 \\ t_5^2 \\ t_5 \\ 1 \end{bmatrix} = [0], \tag{C.12}$$

where E' is a 6×9 matrix whose elements are linear combinations of $s\theta_3$, $c\theta_3$, and 1. Substituting Eqs. (C.10) and (C.11) for $i = 3$ into Eq. (C.12) and multiplying the first four resulting equations by $(1 + t_3^2)$, we obtain

$$E'' \begin{bmatrix} t_4^2 t_5^2 \\ t_4^2 t_5 \\ t_4^2 \\ t_4 t_5^2 \\ t_4 t_5 \\ t_4 \\ t_5^2 \\ t_5 \\ 1 \end{bmatrix} = [0], \tag{C.13}$$

where E'' is a 6×9 matrix. Note that the elements in the first four rows of E'' are quadratic in t_3, whereas the elements in the last two rows are rational functions of t_3, the numerators being quadratic polynomials in t_3 and the denominators being $(1 + t_3^2)$. Multiplying Eq. (C.13) by t_4 yields the following six additional linearly independent equations:

$$E'' \begin{bmatrix} t_4^3 t_5^2 \\ t_4^3 t_5 \\ t_4^3 \\ t_4^2 t_5^2 \\ t_4^2 t_5 \\ t_4^2 \\ t_4 t_5^2 \\ t_4 t_5 \\ t_4 \end{bmatrix} = [0]. \tag{C.14}$$

Finally, we combine Eqs. (C.13) and (C.14) in matrix form:

$$\begin{bmatrix} E'' & 0 \\ 0 & E'' \end{bmatrix} \begin{bmatrix} t_4^3 t_5^2 \\ t_4^3 t_5 \\ t_4^3 \\ t_4^2 t_5^2 \\ t_4^2 t_5 \\ t_4^2 \\ t_4 t_5^2 \\ t_4 t_5 \\ t_4 \\ t_5^2 \\ t_5 \\ 1 \end{bmatrix} = [0]. \tag{C.15}$$

We may consider $t_4^3 t_5^2$, $t_4^3 t_5$, t_4^3, $t_4^2 t_5^2$, $t_4^2 t_5$, t_4^2, $t_4 t_5^2$, $t_4 t_5$, t_4, t_5^2, t_5, and 1 as 12 unknowns. Then Eq. (C.15) constitutes a set of 12 linearly independent equations. The compatibility condition for nontrivial solution to exist is that the coefficient matrix must be singular. Setting the determinant of the coefficient matrix to zero yields a 16th-degree polynomial in t_3. See Raghavan and Roth (1990a) for a more detailed derivation of the equation. Once θ_3 is solved, the other variables can be solved by back substitution. We conclude that the inverse kinematics of the general $6R$ robot has at most 16 real solutions.

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APPENDIX D

LIST OF SYMBOLS

In this appendix we summarize some commonly used symbols. In general, vectors are represented by boldface lowercase characters and matrices are represented by uppercase characters. A trailing superscript denotes the inverse or transpose of a matrix. A trailing subscript may represent the component of a vector, the link number, or the position number of a rigid body. A leading superscript denotes a frame in which a vector or a screw is expressed. The leading superscript is often omitted when the fixed frame serves as the reference frame. In each case, a symbol is defined in the text where it first appears.

A: fixed frame.

A: transpose of a structure matrix, $A = B^{\mathrm{T}}$.

A: $m \times n$ matrix.

A': $m \times n$ matrix.

A, B, C: coefficients of a polynomial equation.

A', B', C': coefficients of a polynomial equation.

A_i: 4×4 matrix of transformation associated with the ith screw displacement.

A_i: ith ball point on the fixed base of a parallel manipulator.

${}^{i-1}A_i$: 4×4 matrix of transformation between the ith and $(i-1)$th link frames.

^0A_n: 4×4 matrix describing the position and orientation of an end effector (link n) with respect to a fixed frame (link 0).

a: link length.

a_i: offset distance between two adjacent joint axes (D-H parameter).

a_i: coefficient of a polynomial system.

a_i^0: complex coefficient of an initial polynomial system.

a_{ij}: (i, j) element of a transformation matrix.

$\mathbf{a}$: vector of constants, $\mathbf{a} = [a_1, a_2, \ldots, a_k]^{\mathrm{T}}$.

$\mathbf{a}^0$: vector of constants, $\mathbf{a}^0 = [a_1^0, a_2^0, \ldots, a_k^0]^{\mathrm{T}}$.

$\mathbf{a}_i$: position vector of a ball point A_i.

B: moving link frame.

B: structure matrix that transforms a set of actuator torques to joint torques.

B: $m \times n$ matrix.

B_i: ith ball point on the moving platform of a parallel manipulator.

B': $m \times n$ matrix.

B^*: structure matrix whose elements assume the value $+1, -1$, or 0.

B^+: pseudoinverse of B, $B^+ = B^{\mathrm{T}}\,[BB^{\mathrm{T}}]^{-1}$.

b: link length.

b_i: damping coefficient.

b_i: random complex constant.

b_i^0: complex constant of an initial polynomial system.

b_{ij}: complex constant.

$\mathbf{b}^0$: vector of constants, $\mathbf{b}^0 = [b_1^0, b_2^0, \cdots, b_n^0]^{\mathrm{T}}$.

$\mathbf{b}_i$: position vector of a ball point B_i relative to a moving frame B, expressed in a fixed frame A.

$^B\mathbf{b}_i$: position vector of a point B_i relative to and expressed in a moving frame B.

C: cylindrical joint.

C: compliance matrix.

C_k: connectivity number associated with limb k.

C_p: cam pair.

c: link length.

c_i: degrees of constraint on relative motion imposed by joint i.

$c\theta_i$: shorthand notation for $\cos\theta_i$.

$\mathbf{c}$: couple.

d: total degree of a polynomial system.

d_i: translational distance along the ith joint axis (D-H parameter).

d_i: ith limb length.

d_i: degree of equation i.

d_{ij}: degree of equation i with respect to the variables of group j.

$d(\cdot)$: derivative of $(\cdot)$.

$\mathbf{d}_i$: vector of a limb i, $\mathbf{d}_i = \overline{A_i B_i}$.

$\det(\cdot)$: determinant of $(\cdot)$.

E: planar pair.

E, E', E'': $m \times n$ matrices.

e: link length.

e_i: link length i.

e_{ij}: coefficient of a polynomial equation.

$\mathbf{e}$: unit vector.

$\mathbf{e}_i$: unit vector directed along the ith joint axis.

F: degrees of freedom of a mechanism or mechanical manipulator.

$F(\mathbf{x})$: system of polynomial equations that are functions of $\mathbf{x}$.

$\mathbf{F}$: end-effector output wrench.

$\mathbf{F}_e$: external wrench exerted at the center of mass on an end effector.

$\mathbf{F}_{i,i-1}$: wrench exerted on link i by link $i-1$.

$\hat{\mathbf{F}}_i$: sum of applied and inertia wrenches exerted at the center of mass of link i.

$\hat{\mathbf{F}}_{ji}$: sum of applied and inertia wrenches exerted at the center of mass of link j, which is a part of limb i.

$\hat{\mathbf{F}}_p$: sum of applied and inertia wrenches exerted at the center of mass of a moving platform.

f_i: degrees of freedom associated with joint i.

f_i: ith polynomial equation of $F(\mathbf{x})$.

$\mathbf{f}$: end-effector output force.

$\mathbf{f}$: system of constraint equations.

$\mathbf{f}^C$: resultant force exerted at the center of mass C of a rigid body.

$\mathbf{f}^O$: resultant force exerted at point O of a rigid body.

$\mathbf{f}_{ai}$: force exerted at point A_i of limb i.

$\mathbf{f}_{bi}$: force exerted at point B_i of a moving platform by limb i.

${}^B\mathbf{f}_{bi}$: $\mathbf{f}_{bi}$ expressed in a moving frame B.

${}^i\mathbf{f}_{bi}$: $\mathbf{f}_{bi}$ expressed in the ith limb frame.

$\mathbf{f}_i$: resultant force exerted at the center of mass of link i.

$\mathbf{f}_i^*$: inertia force exerted at the center of mass of link i, $\mathbf{f}_i^* = -m_i \dot{\mathbf{v}}_i$.

$\hat{\mathbf{f}}_i$: sum of applied and inertia forces exerted at the center of mass of link i, $\hat{\mathbf{f}}_i = \mathbf{f}_i + \mathbf{f}_i^*$.

${}^i\mathbf{f}_i$: $\mathbf{f}_i$ expressed in the ith link frame.

${}^i\mathbf{f}_i^*$: $\mathbf{f}_i^*$ expressed in the ith link frame.

$\mathbf{f}_{i,j}$: force exerted on link i by link j.

${}^k\mathbf{f}_{i,j}$: $\mathbf{f}_{i,j}$ expressed in the kth link frame.

$\mathbf{f}_{ji}^*$: inertia force exerted at the center of mass of link j, which is a part of limb i.

${}^i\mathbf{f}_{ji}^*$: $\mathbf{f}_{ji}^*$ expressed in the ith limb frame.

$\mathbf{f}_p$: resultant force exerted at the center of mass of a moving platform.

$\mathbf{f}_p^*$: inertia force exerted at the center of mass of a moving platform, $\mathbf{f}_p^* = -m_p \dot{\mathbf{v}}_p$.

$\hat{\mathbf{f}}_p$: sum of applied and inertia forces exerted at the center of mass of a moving platform, $\hat{\mathbf{f}}_p = \mathbf{f}_p + \mathbf{f}_p^*$.

$\mathbf{f}_r$: vector of friction torques and/or forces.

G: gear pair.

$G(\mathbf{x})$: initial system of polynomial equations used by a continuation method.

G_i: ith component of $\mathbf{G}$.

${}^{i-1}G_i$: 4×4 transformation matrix that contains only link parameters.

$\mathbf{G}$: vector of gravitational effects, $\mathbf{G} = [G_1, G_2, \ldots, G_n]^{\mathrm{T}}$.

$\tilde{\mathbf{G}}$: $\mathbf{G}$ expressed in the end-effector space.
g: link length.
g_c: gravitational constant.
$g_i(\mathbf{x})$: ith polynomial equation of $G(\mathbf{x})$.
$\mathbf{g}$: acceleration of gravity.
${}^i\mathbf{g}$: $\mathbf{g}$ expressed in the ith link frame.
H: helical joint.
H: $m \times (m-n)$ matrix whose column vectors span the null space of a structure matrix B.
$H(\mathbf{x}, t)$: homotopy function.
${}^{i-1}H_i$: 4×4 transformation matrix that contains only one joint variable.
h: link length.
$h_{ij}(\mathbf{x})$: polynomial of degree d_{ij} with respect to the variables of group j.
$\mathbf{h}_i^A$: angular momentum of limb i taken about point A_i.
${}^i\mathbf{h}_i^A$: $\mathbf{h}_i^A$ expressed in the ith limb frame.
$\mathbf{h}^C$: angular momentum of a rigid body taken about the center of mass C.
$\mathbf{h}_{ji}^C$: angular momentum of link j of the ith limb taken about the center of mass of link j.
$\mathbf{h}^O$: angular momentum of a rigid body taken about point O.
I: identity matrix.
I_B^C: shorthand notation for ${}^AI_B^C$.
I_B^O: shorthand notation for ${}^AI_B^O$.
${}^AI_B^C$: inertia matrix of a rigid body B taken about the center of mass and expressed in a fixed frame A.
${}^AI_B^O$: inertia matrix of a rigid body B taken about a point O and expressed in a fixed frame A.
I_i: inertia matrix of link i taken about the center of mass and expressed in a fixed frame A.
${}^iI_i$: inertia matrix of link i taken about the center of mass and expressed in the ith link frame.
${}^iI_{ji}$: inertia matrix of link j taken about the center of mass link j and expressed in the ith limb frame.
$I_{j,x}, I_{j,y}, I_{j,z}$: principal moment of inertia of link j about the coordinate axes of a fixed frame.

$I_{j,u}, I_{j,v}, I_{j,w}$: principal moment of inertia of link j about the coordinate axes of a moving frame.

I_{xx}, I_{yy}, I_{zz}: principal moment of inertia of a rigid body about the coordinate axes of a fixed frame.

I_{xy}, I_{yz}, I_{zx}: product of inertia of a rigid body about the coordinate axes of a fixed frame.

I_{uu}, I_{vv}, I_{ww}: principal moment of inertia of a rigid body about the coordinate axes of a moving frame.

I_{uv}, I_{vw}, I_{wu}: product of inertia of a rigid body about the coordinate axes of a moving frame.

I_{uv}^O: product of inertia of a rigid body relative to a reference point O for **u** and **v**.

$\mathbf{I}_u^O$: second moment of inertia of a rigid body relative to a line that passes through a point O and is parallel to a unit vector **u**.

i: unit vector pointing along the x-axis in a Cartesian coordinate system.

J: Jacobian matrix.

J_{bi}: 3×6 matrix transforming the velocity of the center of mass of a moving platform to a ball point B_i.

$J_{bi,x}$: first row of J_{bi}.

$J_{bi,y}$: second row of J_{bi}.

$J_{bi,z}$: third row of J_{bi}.

J_i: ith link Jacobian matrix.

J_{ji}: jth link Jacobian matrix of the ith limb.

J_p: Jacobian matrix of a moving platform.

J_q: Jacobian matrix associated with the rates of change of the input joint variables.

J_{vi}: link Jacobian submatrix associated with the linear velocity of the center of mass of link i.

J_{vi}^j: jth column of J_{vi}.

$J_{\omega i}$: link Jacobian submatrix associated with the angular velocity vector of link i.

$J_{\omega i}^j$: jth column of $J_{\omega i}$.

J_x: Jacobian matrix associated with the rates of change of the end effector coordinates.

j: number of joints in a mechanism.

j_g: number of gear pairs.

j_i: number of joints with i degrees of freedom.
j_t: number of turning pairs.
$\mathbf{j}$: unit vector pointing along the y-axis in a Cartesian coordinate system.
K: kinetic energy of a system of rigid bodies.
K: stiffness matrix.
K_j: kinetic energy of link j.
K_r: kinetic energy of rotor r.
k_i: constant coefficient.
k_i: stiffness constant associated with joint i.
k_p: position feedback gain.
k_v: velocity feedback gain.
$\mathbf{k}$: unit vector pointing along the z-axis in a Cartesian coordinate system.
L: Lagrangian function.
L: number of independent loops in a mechanism.
L, M, N, P, Q, R: Plücker coordinates of a line.
$\mathbf{l}^O$: linear momentum of a rigid body about a reference point O.
M: manipulator inertia matrix.
$\tilde{M}$: M expressed in the end-effector space.
M_{ij}: (i, j) element of M.
m: mass of a rigid body.
m: number of tendons.
m_i: mass of link i.
$\mathbf{m}_i$: unit vector.
N_{ij}: gear ratio between a gear pair i and j, $N_{ij} = T_j/T_i$.
n: number of links in a mechanism (including the fixed link).
n: number of degrees of freedom of a manipulator.
n: last link of a manipulator.
n: number of generalized coordinates.
$\mathbf{n}$: end effector output moment.
$\mathbf{n}^C$: resulting moment exerted at the center of mass C of a rigid body.
$\mathbf{n}^O$: resulting moment taken about point O of a rigid body.

$\mathbf{n}_i$: unit vector.

$\mathbf{n}_i$: resulting moment exerted at the center of mass of link i.

$\hat{\mathbf{n}}_i$: sum of applied and inertia moments exerted at the center of mass of link i, $\hat{\mathbf{n}}_i = \mathbf{n}_i + \mathbf{n}_i^*$.

$\mathbf{n}_i^A$: resulting moment taken about point A of link i.

$\mathbf{n}_i^*$: inertia moment exerted at the center of mass of link i, $\mathbf{n}_i^* = -I_i\dot{\boldsymbol{\omega}}_i - \boldsymbol{\omega}_i \times (I_i\boldsymbol{\omega}_i)$.

${}^i\mathbf{n}_i$: $\mathbf{n}_i$ expressed in the ith link frame.

${}^i\mathbf{n}_i^*$: $\mathbf{n}_i^*$ expressed in the ith link frame.

$\mathbf{n}_{i,j}$: moment exerted on link i by link j.

${}^k\mathbf{n}_{i,j}$: $\mathbf{n}_{i,j}$ expressed in the kth link frame.

$\mathbf{n}_{ji}^*$: inertia moment exerted at the center of mass of link j which is a part of limb i.

${}^i\mathbf{n}_{ji}^*$: $\mathbf{n}_{ji}^*$ expressed in the ith limb frame.

$\mathbf{n}_p$: resulting moment exerted at the center of mass of a moving platform.

${}^B\mathbf{n}_p$: $\mathbf{n}_p$ expressed in a moving frame B.

$\mathbf{n}_p^*$: inertia moment exerted at the center of mass of a moving platform, $\mathbf{n}_p^* = -I_p\dot{\boldsymbol{\omega}}_p - \boldsymbol{\omega}_p \times (I_p\boldsymbol{\omega}_p)$.

$\hat{\mathbf{n}}_p$: sum of applied and inertia moments exerted at the center of mass of a moving platform, $\hat{\mathbf{n}}_p = \mathbf{n}_p + \mathbf{n}_p^*$.

O: origin of a fixed frame, $O = O_0$.

$O^+(x)$: "+" operator that takes the value of x when $x \geq 0$, and 0 otherwise.

$O^-(x)$: "−" operator that takes the value of $-x$ when $x < 0$, and 0 otherwise.

O_i: origin of the ith link frame.

P: point of a rigid body.

P: point located at a wrist center of a serial manipulator.

P: point located at the center of mass of a moving platform.

p_x, p_y, p_z: x, y, and z coordinates of $\mathbf{p}$.

p_u, p_v, p_w: u, v, and w coordinates of $\mathbf{p}$.

$\mathbf{p}$: abbreviated form of ${}^A\mathbf{p}$.

${}^A\mathbf{p}$: position vector of a point P with respect to a fixed frame A.

${}^A\mathbf{p}$: position vector of the wrist center with respect to a fixed frame A.

${}^A\mathbf{p}$: position vector of the center of mass of a moving platform with respect to a fixed frame A.

${}^B\mathbf{p}$: position vector of a point P with respect to a moving frame B.

${}^j\mathbf{p}$: position vector of a point P relative to the jth link frame.

$\mathbf{p}_c$: position vector of the center of mass of a rigid body.

$\mathbf{p}_{ci}$: position vector of the center of mass of link i with respect to a fixed frame.

${}^j\mathbf{p}_{ci}$: position vector of the center of mass of link i with respect to the jth link frame.

${}^j\mathbf{p}^*_{ci}$: position vector of the center of mass of link i with respect to the jth link frame, expressed in a fixed frame.

$\mathbf{p}_i$: position vector of the origin O_i of the ith link frame with respect to a fixed frame.

${}^j\mathbf{p}_i$: $\overline{O_j O_i}$ expressed in the jth link frame.

${}^j\mathbf{p}^*_i$: $\overline{O_j O_i}$ expressed in a fixed frame.

${}^j\mathbf{p}_n$: $\overline{O_j O_n}$ expressed in the jth link frame.

${}^j\mathbf{p}^*_n$: $\overline{O_j O_n}$ expressed in a fixed frame.

Q: origin of an end-effector coordinate frame.

Q_i: generalized active force associated with the ith generalized coordinate.

$\mathbf{Q}$: vector of generalized forces, $\mathbf{Q} = [Q_1, Q_2, \cdots, Q_n]^{\mathrm{T}}$.

$\tilde{\mathbf{Q}}$: $\mathbf{Q}$ expressed in the end-effector space.

q_i: ith joint variable or generalized coordinate.

$\mathbf{q}$: vector of joint angles or generalized coordinates, $\mathbf{q} = [q_1, q_2, \ldots, q_n]^{\mathrm{T}}$.

$\mathbf{q}$: abbreviated form of ${}^A\mathbf{q}$.

${}^A\mathbf{q}$: position vector of the origin of a moving frame with respect to a fixed frame A.

${}^A\mathbf{q}$: position vector of the end-effector coordinate frame with respect to a fixed frame A.

R: revolute joint.

R^*: $n \times n$ diagonal matrix whose nonzero elements represent the radii of the pulleys installed on the joint axes of a tendon-driven manipulator.

${}^A R_B$: 3×3 rotation matrix that describes the orientation of frame B with respect to frame A.

${}^B R_A$: inverse transformation of ${}^A R_B$, ${}^B R_A = {}^A R_B^{-1}$.

r: link length.

r_a: radius of gyration.

r_i: radius of pulley i.

$\mathbf{r}$: position vector of a point relative to a center-of-mass coordinate frame.

$\mathbf{r}_{ci}$: position vector of the center of mass of link i relative to the ith link frame, expressed in a fixed frame.

${}^i\mathbf{r}_{ci}$: $\mathbf{r}_{ci}$ expressed in the ith link frame.

$\mathbf{r}_i$: $\overline{O_{i-1} O_i}$ expressed in a fixed frame.

${}^i\mathbf{r}_i$: $\mathbf{r}_i$ expressed in link frame i, ${}^i\mathbf{r}_i = [a_i, d_i s\alpha, d_i c\alpha]^T$.

S: spherical joint.

S_i: ith coordinate of a screw \$.

S_{ri}: ith coordinate of a reciprocal screw $\$_r$.

s_i: linear displacement of tendon i.

$\mathbf{s}$: vector of tendon displacements, $\mathbf{s} = [s_1, s_2, \ldots, s_m]^T$.

$\mathbf{s}_i$: unit vector pointing along the axis of a screw $\$_i$.

${}^i\mathbf{s}_i$: $\mathbf{s}_i$ expressed in the ith limb frame.

$\mathbf{s}_o$: position vector of a screw axis defined from the origin of a fixed frame to any point on the screw axis.

$\mathbf{s}_{oi}$: position vector of a screw axis defined from the origin of a fixed frame to any point on the ith screw axis.

${}^j\mathbf{s}_{oi}$: position vector of a screw axis defined from the origin of a reference frame j to any point on the screw axis i.

$\mathbf{s}_r$: unit vector pointing along the axis of a reciprocal screw $\$_r$.

$\mathbf{s}_{ro}$: position vector of a screw axis defined from the origin of a fixed frame to any point on the reciprocal screw $\$_r$.

$s\theta_i$: shorthand notation for $\sin\theta_i$.

T_i: number of teeth on gear i.

^AT_B: 4×4 matrix describing the transformation of frame B to frame A.

$^i\tilde{T}_j$: 6×6 matrix describing the transformation of a screw or wrench from coordinate frame j to i.

t: time.

t: continuation parameter.

t: translational distance along the axis of a screw.

t_i: translational distance along the ith screw axis.

t_i: shorthand notation for $\tan(\theta_i/2)$.

U: potential energy of a mechanical system.

U_i: potential energy of link i.

u_x, u_y, u_z: x, y, and z components of $\mathbf{u}$.

$\mathbf{u}$: unit vector pointing along the u-axis of a moving frame.

V_i: ith component of $\mathbf{V}$.

$\mathbf{V}$: vector of velocity coupling terms, $\mathbf{V} = [V_1, V_2, \ldots, V_n]^{\mathrm{T}}$.

$\tilde{\mathbf{V}}$: $\mathbf{V}$ expressed in the end-effector space.

v_x, v_y, v_z: x, y, and z components of $\mathbf{v}$.

$\mathbf{v}$: unit vector pointing along the v-axis of a moving frame.

$\mathbf{v}$: abbreviated form of $^A\mathbf{v}$.

$^A\mathbf{v}$: vector $\mathbf{v}$ expressed in a fixed frame A.

$\mathbf{v}_{bi}$: velocity of a ball point B_i expressed in a fixed frame.

$^i\mathbf{v}_{bi}$: $\mathbf{v}_{bi}$ expressed in the ith limb frame.

$\mathbf{v}_{ci}$: velocity of the center of mass of link i relative to a fixed frame.

$^j\mathbf{v}_{ci}$: $\mathbf{v}_{ci}$ expressed in the jth link frame.

$\dot{\mathbf{v}}_{ci}$: linear acceleration of the center of mass of link i relative to a fixed frame.

$^j\dot{\mathbf{v}}_{ci}$: $\dot{\mathbf{v}}_{ci}$ expressed in the jth link frame.

$\mathbf{v}_i$: velocity of the origin O_i of the ith link frame relative to a fixed frame.

$\dot{\mathbf{v}}_i$: linear acceleration of the origin O_i of the ith link frame relative to a fixed frame.

$^i\mathbf{v}_i$: $\mathbf{v}_i$ expressed in the ith link frame.

$\mathbf{v}_{ji}$: velocity of the center of mass of link j, which is a part of limb i.

$\mathbf{v}_n$: velocity of the origin of an end-effector coordinate frame.

$\mathbf{v}_o$: velocity of a point in the end effector which instantaneously coincides with the origin of a fixed frame.

$\mathbf{v}_p$: velocity of a point P relative to a fixed frame.

$\mathbf{v}_p$: velocity of the center of mass of a moving platform relative to a fixed frame.

$\dot{\mathbf{v}}_p$: acceleration of a point P relative to a fixed frame.

$\dot{\mathbf{v}}_p$: acceleration of the center of mass of a moving platform relative to a fixed frame.

$\mathbf{v}_q$: velocity of a point Q relative to a fixed frame.

${}^iW_j$: 3×3 skew-symmetric matrix whose nonzero elements represent the vector $\overline{O_i O_j}$ expressed in the ith link frame.

w_x, w_y, w_z: x, y, and z components of $\mathbf{w}$.

$\mathbf{w}$: unit vector pointing along the w-axis of a moving frame.

x_c, y_c, z_c: x, y, and z axes of a coordinate frame attached to the center of mass of link i.

x_i: ith variable of a polynomial system.

x_i, y_i, z_i: x, y, and z axes of the ith link frame.

$\mathbf{x}$: n-dimensional vector describing the position and orientation of an end effector.

$\dot{\mathbf{x}}$: n-dimensional vector describing the linear and angular velocities of an end effector, $\dot{\mathbf{x}}_i = [v_x, v_y, \ldots, \omega_z]^T$ for conventional Jacobian and $\dot{\mathbf{x}}_i = [\omega_x, \omega_y, \ldots, v_z]^T$ for screw-based Jacobian.

$\mathbf{x}_i$: n-dimensional vector describing the position and orientation of link i.

$\dot{\mathbf{x}}_i$: n-dimensional vector describing the linear and angular velocities of link i.

$\dot{\mathbf{x}}_{ji}$: n-dimensional vector describing the linear and angular velocities of the jth link, which is a part of limb i.

$\dot{\mathbf{x}}_p$: n-dimensional vector describing the linear and angular velocities of a moving platform.

y_i: ith variable of a polynomial system.

${}^{i-1}Z_{i-1}$: 3×3 skew-symmetric matrix whose nonzero elements represent a unit angular velocity of link i relative to link $i-1$, expressed in the $(i-1)$th link frame.

$\mathbf{z}_i$: unit vector pointing along the z_i axis, expressed in a fixed frame.

${}^i\mathbf{z}_i$: $\mathbf{z}_i$ expressed in the ith link frame, ${}^i\mathbf{z}_i = [0, 0, 1]^T$.

α: angle.

α_i: twist angle between two adjacent joint axes of link i (D-H parameter).

β: angle.

β_i: variable.

Γ_i: ith constraint function.

γ: random complex constant.

$\boldsymbol{\gamma}$: 1×3 perspective transformation matrix.

$\Delta\mathbf{q}$: vector of infinitesimal joint displacements.

$\Delta\mathbf{x}$: vector of infinitesimal end-effector displacement.

δ: bias force.

δ_i: ith bias force.

$\delta\mathbf{x}_i$: virtual displacement of link i.

$\delta\mathbf{x}_p$: virtual displacement of a moving platform.

δW: virtual work.

η: virtual coefficient.

θ: angle.

θ: Euler angle.

θ_i: ith joint angle (D-H parameter).

θ_{ij}: shorthand notation for $\theta_i + \theta_j$.

θ_{ijk}: shorthand notation for $\theta_i + \theta_j + \theta_k$.

$\theta_{i,j}$: angular displacement of link i relative to link j.

θ_{ji}: angular displacement associated with the jth joint of limb i.

$\boldsymbol{\theta}$: vector of joint angles.

κ_i: ith constant coefficient of an equation.

λ: degrees of freedom of the space in which a mechanism is intended to function.

λ: eigenvalue.

λ: constant.

λ: pitch of a screw.

λ_i: ith Lagrange multiplier.

$\boldsymbol{\lambda}$: vector of some constants.

μ: constant.

μ_i: ith constant.

ν: constant.

ν_i: ith constant.

ξ_i: ith tendon force.

ξ_i: ith actuator torque or force.

$\boldsymbol{\xi}$: vector of tendon forces.

$\boldsymbol{\xi}$: vector of actuator torques and/or forces.

ρ: scaling factor.

ρ: material density.

τ_i: ith joint torque or force.

$\boldsymbol{\tau}$: vector of joint torques and/or forces.

ϕ: angle.

ϕ: Euler angle.

ϕ_i: angle i.

$\boldsymbol{\phi}$: vector of actuator rotation angles.

χ: $n \times n$ diagonal matrix whose elements represent the stiffness constants of actuated joints.

ψ: angle.

ψ: Euler angle.

ψ_i: angle i.

Ω: 3×3 skew symmetric matrix whose nonzero elements represent the angular velocity of a rigid body, $\Omega = {}^A\dot{R}_B\, {}^AR_B^{\mathrm{T}}$.

$\omega_x, \omega_y, \omega_z$: x, y, and z components of $\boldsymbol{\omega}_i$.

$\boldsymbol{\omega}$: angular velocity of a rigid body.

${}^A\boldsymbol{\omega}_B$: angular velocity of a moving frame B relative to a fixed frame A.

$\boldsymbol{\omega}_i$: angular velocity of link i.

$\dot{\boldsymbol{\omega}}_i$: angular acceleration of link i.

${}^j\boldsymbol{\omega}_i$: $\boldsymbol{\omega}_i$ expressed in the jth link frame.

${}^j\dot{\boldsymbol{\omega}}_i$: $\dot{\boldsymbol{\omega}}_i$ expressed in the jth link frame.

$\boldsymbol{\omega}_p$: angular velocity of a moving platform.

$\dot{\boldsymbol{\omega}}_p$: angular acceleration of a moving platform.

$\$$: screw.

$\hat{\$}$: unit screw.

${}^{j}\$$: screw expressed in the jth link frame.

$\$_{\mathbf{i}}$: screw displacement associated with the ith joint axis.

$\hat{\$}_{j,i}$: unit screw associated with the jth joint of the ith limb.

$\$_{p}$: screw displacement associated with a moving platform.

$\$_{r}$: reciprocal screw.

$\hat{\$}_{r}$: unit reciprocal screw.

$\prod$: product convention.

$\sum$: summation convention.

$\int$: integration convention.

${}^{k}(\cdot)$: $(\cdot)$ expressed in the kth link frame.

$(\cdot)^{\mathrm{T}}$: transpose of $(\cdot)$.

$(\cdot)^{-1}$: inverse of $(\cdot)$.

$(\dot{\cdot})$: derivative of $(\cdot)$ with respect to time.

$|(\cdot)|$: absolute value of $(\cdot)$.

$|(\cdot)|$: determinant of $(\cdot)$.

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